Software Testing & Quality Assurance

Regression Testing

Roadmap

Beizer's Maturity Model

Test Automation

Unit Testing

Control/Data-Flow Analysis

Code Coverage

Mutation Testing

Regression Testing

Fuzzing

Search-based Test Generation

Model-Based Testing

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Model-Based Testing

Motivation

Changes have been made to the system's code

…or to the system's environment

Have new faults been introduced that weren't there before?

Can we **confidently** deploy the new version of the system?

Regression Testing is now standard in the software development industry

Typically based on a **regression test suite**

Scenarios

Maven™

Regression testing can be **very costly**

- As software grows, so do test suites
- **CI/CD** (Continuous Integration & Development) frameworks help
- Running regression test suites overnight is common practice

Rerunning existing tests **may not always be possible**, e.g. if public interfaces have changed

For embedded systems, e.g. car controllers, retesting the software **in use** may take **months**. Simulation helps but is not a complete solution.

Regression Testing Techniques

Minimisation – Could my test suite be smaller?

Typically based on a criterion, i.e., minimise while retaining ...

Prioritisation – In which order should I run my tests?

To detect new defects as soon as possible

Selection – Which tests should I run?

Given a set of lines of code

Minimisation

Minimisation

As software grows, so does the size of its test suite!

- Tests become **obsolete**, or **redundant**
- **Test maintainability** is rarely a priority in practice

Removing redundancy

- Many similar tests covering the same code!
- Coverage information from CI/CD, i.e., tests have been run before
- Keep only tests that execute changed (or deleted) code; deem other tests redundant wrt changes, so no need to re-execute them!

Minimisation Algorithm

Focus on code coverage, e.g., **branch coverage**

Alternative approaches exist, e.g., to preserve other properties

Given a regression test suite , **find the smallest subset** $'$ $\in T$ **such that** $'$ **and** **have the same coverage**

The hope is that preserving coverage leads to us preserving effectiveness, i.e., $'$ is as effective as

Not necessarily **the** smallest test suite: If the cost of test execution varies then we may want a **cheapest-to-execute** test suite

Formalisation

Given a set of coverage goals $C = \{c_1, \ldots, c_k\}$(e.g., lines)

a.k.a. **test requirements**

And a test suite $T = \{t_1, \ldots, t_n\}$that meets/covers all of them

Where each $t_i \in T$ covers a given set C_{t_i} of coverage goals

Find the smallest $' \in T$ such that $\bigcup_{t_i \in T'} C_{t_i} = C$

This corresponds to the **NP-complete Set Cover Problem**

Exact Solutions vs Heuristics

Test suites are usually large – otherwise no need to minimise them!

So, our problem instances are usually large, too

Typically infeasible to solve the problem exactly: we fall back upon heuristics: **How to find a good (small enough) test suite?**

A Simple Greedy Algorithm

Start with the empty set $\emptyset$

Add a test t_i that covers **most goals** and remove it from the test pool

Iterations: add the next test that covers the most goals

Terminate when the test suite provides **full coverage**

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** and **E** (either order)

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** and **E** (either order)

Next, **B** is the obvious choice

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** and **E** (either order)

Next, **B** is the obvious choice

Finally, **C** and **D** (either order)

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** and **E** (either order)

Next, **B** is the obvious choice

Finally, **C** and **D** (either order)

Resulting set: **{A, E, B, C, D}**

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** and **E** (either order)

Next, **B** is the obvious choice

Finally, **C** and **D** (either order)

Resulting set: **{A, E, B, C, D}**

But could have just used {C, D}!

A Simple Greedy Algorithm

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** and **E** (either order)

Next, **B** is the obvious choice

Finally, **C** and **D** (either order)

Resulting set: **{A, E, B, C, D}**

But could have just used {C, D}!

Not always a good approach!

Additional Greedy

Take into account the coverage **already achieved**!

Start again with the empty set Ø

Add one of the tests that covers most goals

Iterations: add a test that covers most **currently uncovered** goals

Terminate with **full coverage**

Effective wrt Set Cover Problem – good approximation

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** or **E** (either one)

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** or **E** (either one)

Uncovered goals: $\mathbf{b_4}$ and $\mathbf{b_5}$

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** or **E** (either one)

Uncovered goals: $\mathbf{b_4}$ and $\mathbf{b_5}$

Add **C** and **D** (either order)

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** or **E** (either one)

- Uncovered goals: $\mathbf{b_4}$ and $\mathbf{b_5}$

Add **C** and **D** (either order)

- Each covers one uncovered goal

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
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D					x	x	x	x
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Starts with **A** or **E** (either one)

Uncovered goals: $\mathbf{b_4}$ and $\mathbf{b_5}$

Add **C** and **D** (either order)

Each covers one uncovered goal

Resulting set: **{A, C, D}**

Additional Greedy

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Starts with **A** or **E** (either one)

Uncovered goals: $\mathbf{b_4}$ and $\mathbf{b_5}$

Add **C** and **D** (either order)

Each covers one uncovered goal

Resulting set: **{A, C, D}**

Not optimal, but better!

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Harrold et al (1993)

Idea: **Uniquely-covered** goals first

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

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C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Idea: **Uniquely-covered** goals first

b_4 uniquely-covered by **C**; so add **C**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Idea: **Uniquely-covered** goals first

b_4 uniquely-covered by **C**; so add **C**

b_5 uniquely-covered by **D**; so add **D**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Idea: **Uniquely-covered** goals first

b_4 uniquely-covered by **C**; so add **C**

b_5 uniquely-covered by **D**; so add **D**

Iterations: consider goals covered by two, three, etc tests

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Idea: **Uniquely-covered** goals first

b_4 uniquely-covered by **C**; so add **C**

b_5 uniquely-covered by **D**; so add **D**

Iterations: consider goals covered by two, three, etc tests

Ties need resolution

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Idea: **Uniquely-covered** goals first

b_4 uniquely-covered by **C**; so add **C**

b_5 uniquely-covered by **D**; so add **D**

Iterations: consider goals covered by two, three, etc tests

Ties need resolution

Terminate with **full coverage**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x
E	x	x	x			x	x	x

Idea: **Uniquely-covered** goals first

b_4 uniquely-covered by **C**; so add **C**

b_5 uniquely-covered by **D**; so add **D**

Iterations: consider goals covered by two, three, etc tests

Ties need resolution

Terminate with **full coverage**

Resulting set: **{C, D}**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

Harrold et al (1993)

b_4 uniquely-covered by **C**; so **add C**

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

b_4 uniquely-covered by **C**; so **add C**

Next, **b_5** is covered by **2** tests: **B**, **D**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

b_4 uniquely-covered by **C**; so **add C**

Next, **b_5** is covered by **2** tests: **B**, **D**

Tie resolution between **B** and **D**:

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

b_4 uniquely-covered by **C**; so **add C**

Next, **b_5** is covered by **2** tests: **B**, **D**

Tie resolution between **B** and **D**:

Look at goals covered by **3** tests

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

b_4 uniquely-covered by **C**; so **add C**

Next, **b_5** is covered by **2** tests: **B**, **D**

Tie resolution between **B** and **D**:

Look at goals covered by **3** tests

B : b_1, b_2, b_3 vs **D : b_6** ; so **add B**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

b_4 uniquely-covered by **C**; so **add C**

Next, **b_5** is covered by **2** tests: **B**, **D**

Tie resolution between **B** and **D**:

Look at goals covered by **3** tests

B : b_1, b_2, b_3 vs **D : b_6** ; so **add B**

Next, **b_6** is covered by **3** tests: **A**, **D**, **E**

Harrold et al (1993)

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x		x		x	x
C	-	-	-	x				
D					x	x	x	x
E	x	x	x			x	x	x

b_4 uniquely-covered by **C**; so **add C**

Next, **b_5** is covered by **2** tests: **B**, **D**

Tie resolution between **B** and **D**:

Look at goals covered by **3** tests

B : b_1, b_2, b_3 vs **D : b_6** ; so **add B**

Next, **b_6** is covered by **3** tests: **A**, **D**, **E**

Resulting set: **{C, B, A|D|E}**

Multi-Objective Approach

Metaheuristics, e.g., Genetic Algorithms

Enough improvements over Greedy to be worthwhile?

Problem generalisation: Find a regression test suite $'$ such that **there is no smaller test suite** that provides the same coverage as $'$.

Optimise two objective functions: **maximise** coverage & **minimise** cost

More flexible than Greedy: other coverage criteria as added targets

Multi-objective optimisation algorithms return a set of solutions with trade-offs

Multi-Objective Approach

Pareto Dominance: Approach to comparing candidate solutions

Given two candidate solutions **x** and **y**, **x** Pareto-dominates **y** if **x** is **at least as good as y** on all objectives and **strictly better** than **y** on at least one objective

- i.e., we would **never** choose **y** over **x**

Ideally, we want to find the **Pareto Front**: the set of solutions that are not Pareto-dominated by any other solution

Many metaheuristic algorithms, the most famous is probably the Non-dominated Sorting Genetic Algorithm II (NSGA-II); there is now NSGA-III

Prioritisation

Prioritisation

Goal: Find a good **test execution order**

To meet test requirements, e.g., coverage, **as early as possible**

Ideally, we would like any test failures to occur **as early as possible**

Speed up software development: **Stop testing once a test fails**

Even if we plan to execute all tests, the sooner we find failures the sooner we can start trying to fix the code

Problem

Problem: We do not know in advance **which tests will fail**!

So, the best order is **unknown**

Idea: use metrics and historical information associated with faults

Prioritise tests that are deemed **more likely** to lead to failures

While quickly **maximising coverage**

Hope: **Find faults early**!

Using Coverage

We might only look at coverage

- Aim to achieve 100% coverage as quickly as possible
- Maximise coverage for a given budget, e.g. number of tests
- Achieve coverage **quickly** to get **good** coverage whenever we stop

Greedy Algorithm

Start with **A**, then choose **D**, then **C**

To achieve 100% coverage **faster**, we should do **C** then **D** or viceversa

Suboptimal if we must stop after just one test

Test\branch	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
A	x	x	x			x	x	x
B	x	x	x				x	x
C	x	x	x	x				
D					x	x	x	x

Using Fault-Detection Data

Using Fault-Detection Data

Use Mutation Testing data!

Using Fault-Detection Data

Use Mutation Testing data!

15 mutants in total

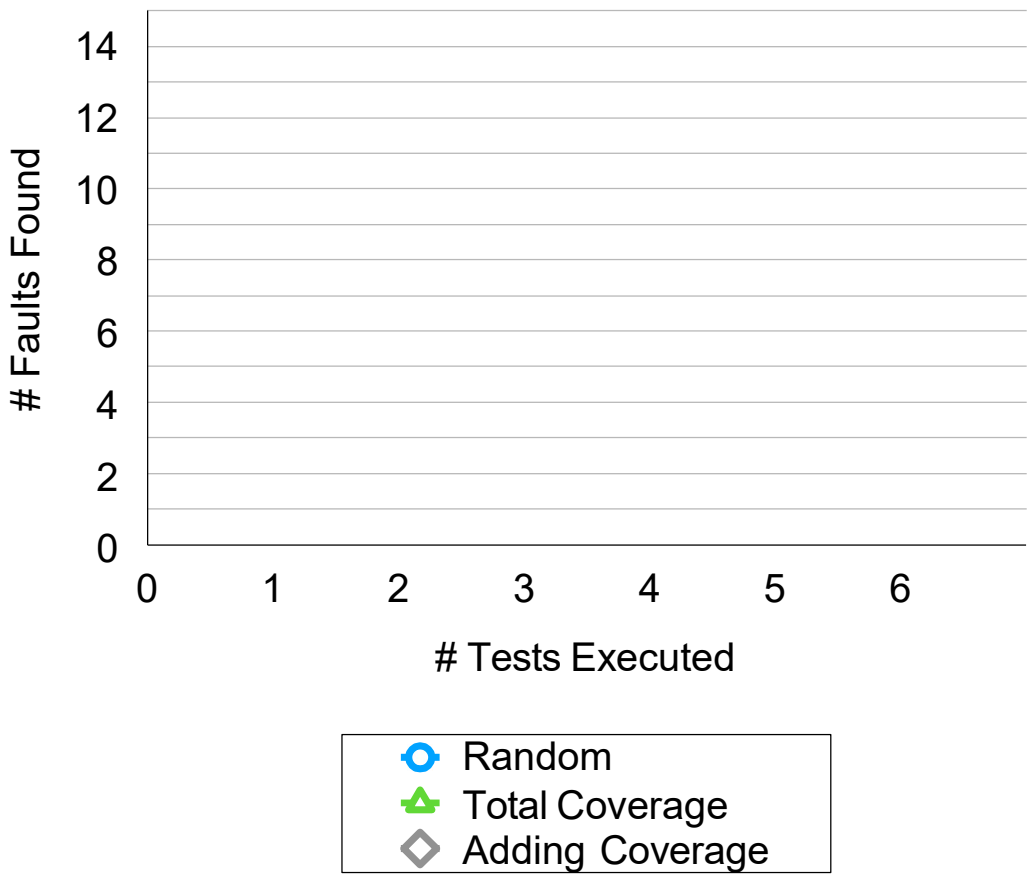
Using Fault-Detection Data

Use Mutation Testing data!

15 mutants in total

Tests A-F kill 13 of them

Using Fault-Detection Data

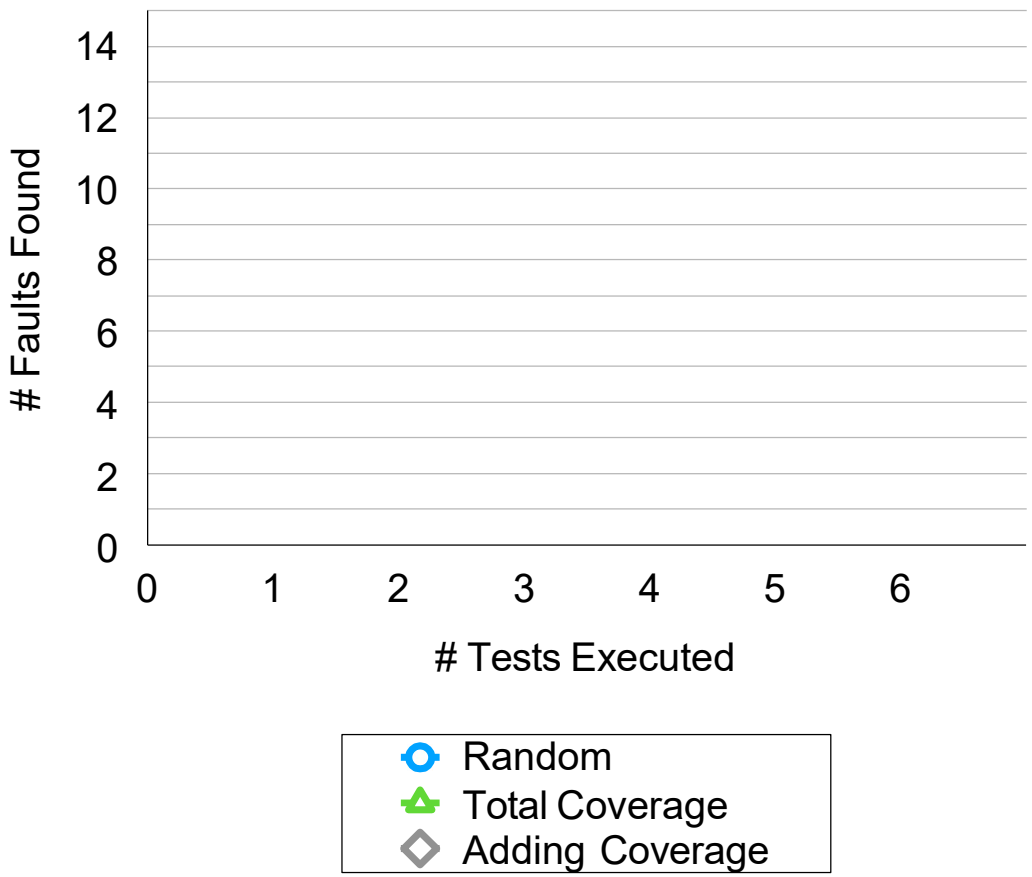


Use Mutation Testing data!

15 mutants in total

Tests A-F kill 13 of them

Using Fault-Detection Data



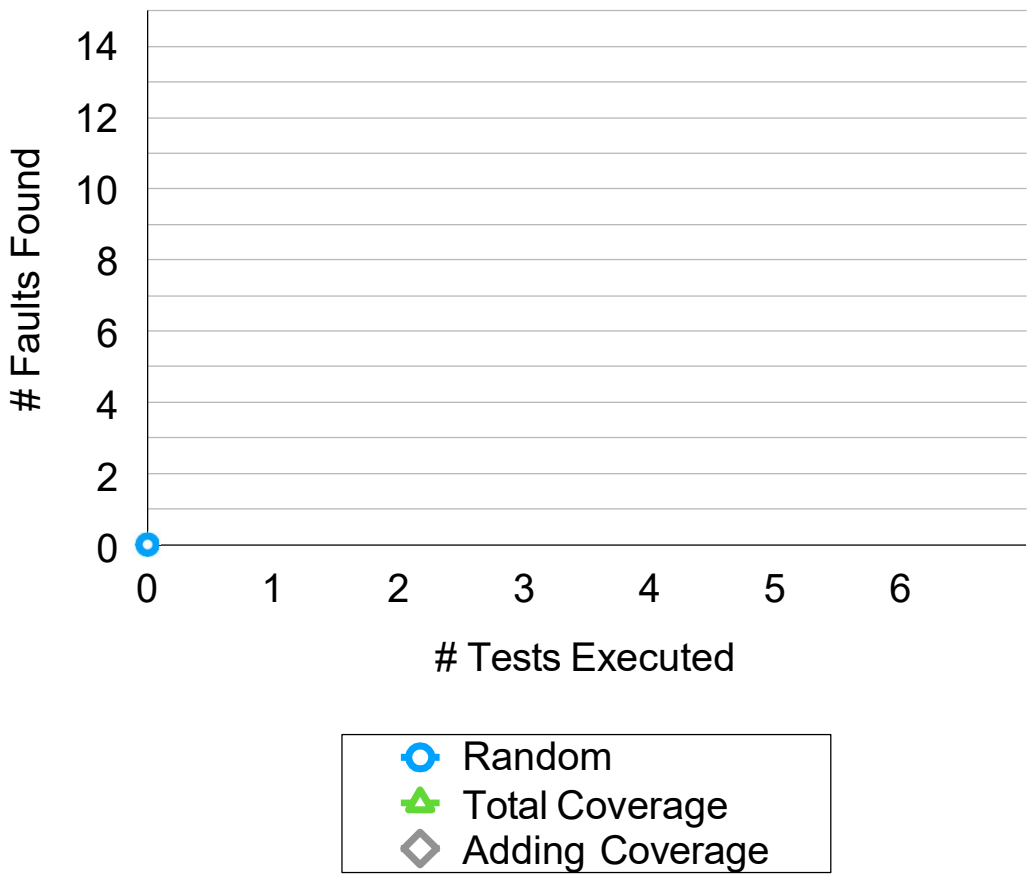
Use Mutation Testing data!

15 mutants in total

Tests A-F kill 13 of them

Random: E-A-B-F-D-C

Using Fault-Detection Data



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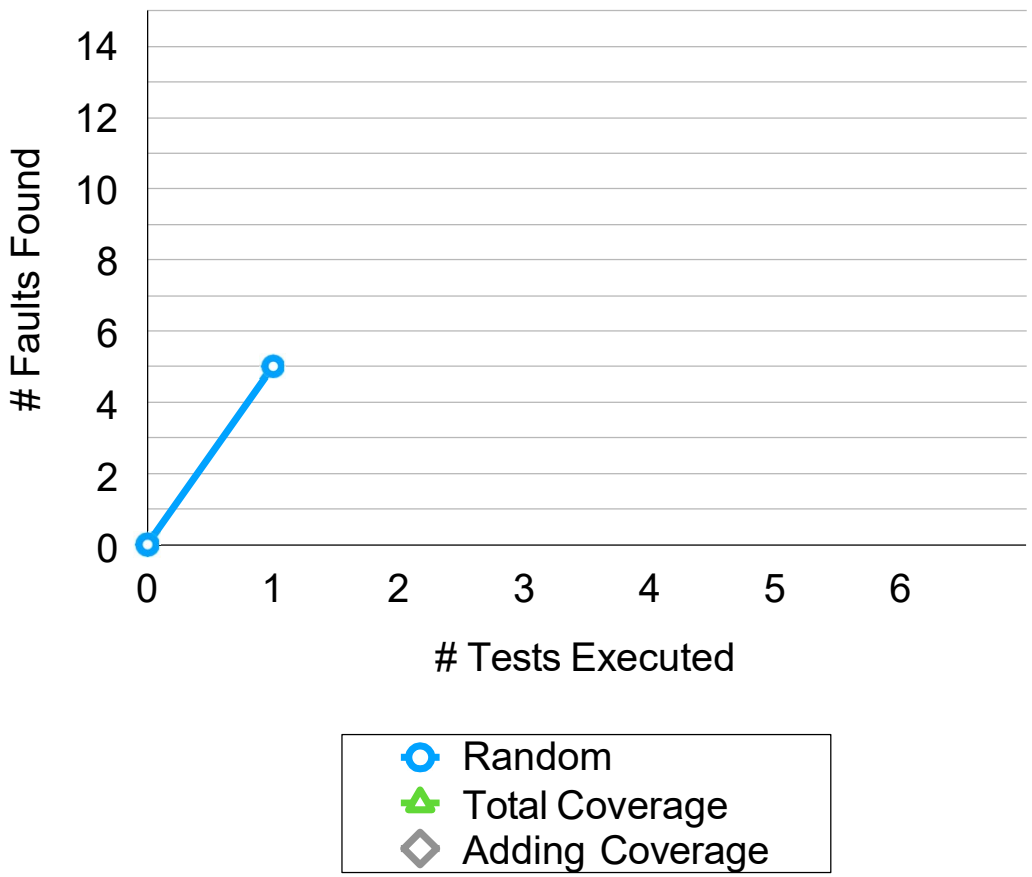
Random: E-A-B-F-D-C

Using Fault-Detection Data



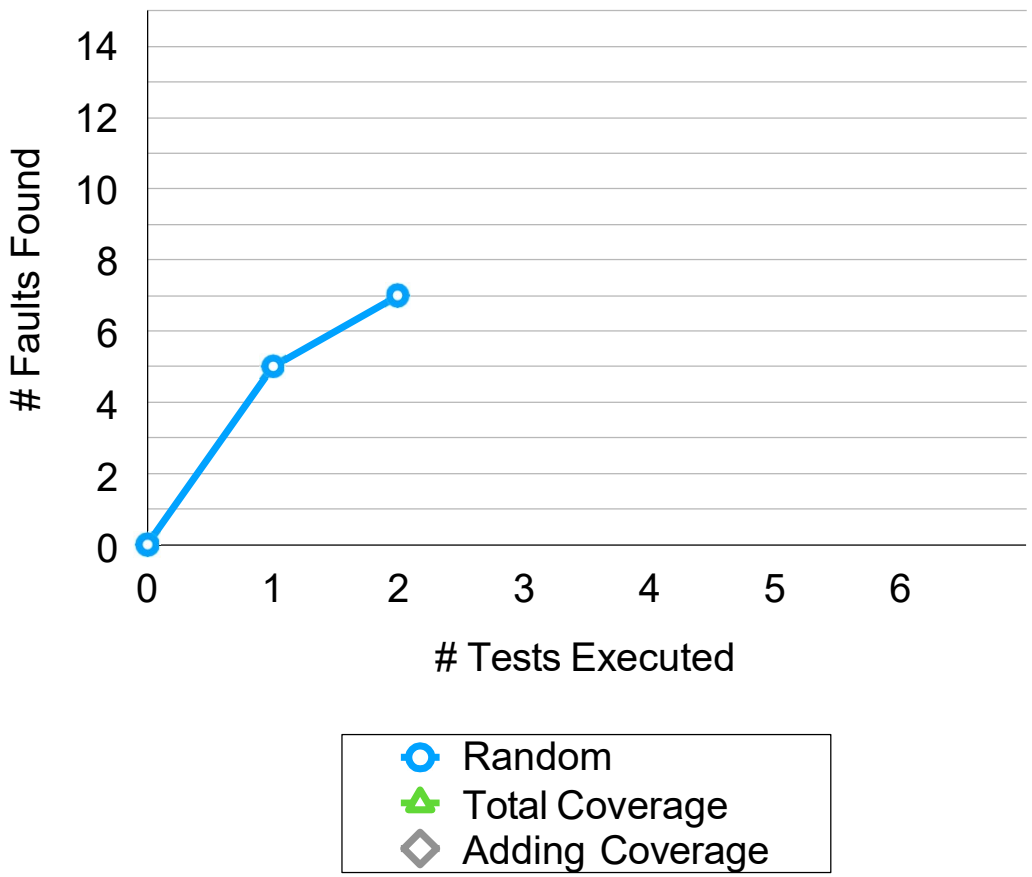
Use Mutation Testing data!

15 mutants in total

Tests A-F kill 13 of them

Random: E-A-B-F-D-C

Using Fault-Detection Data



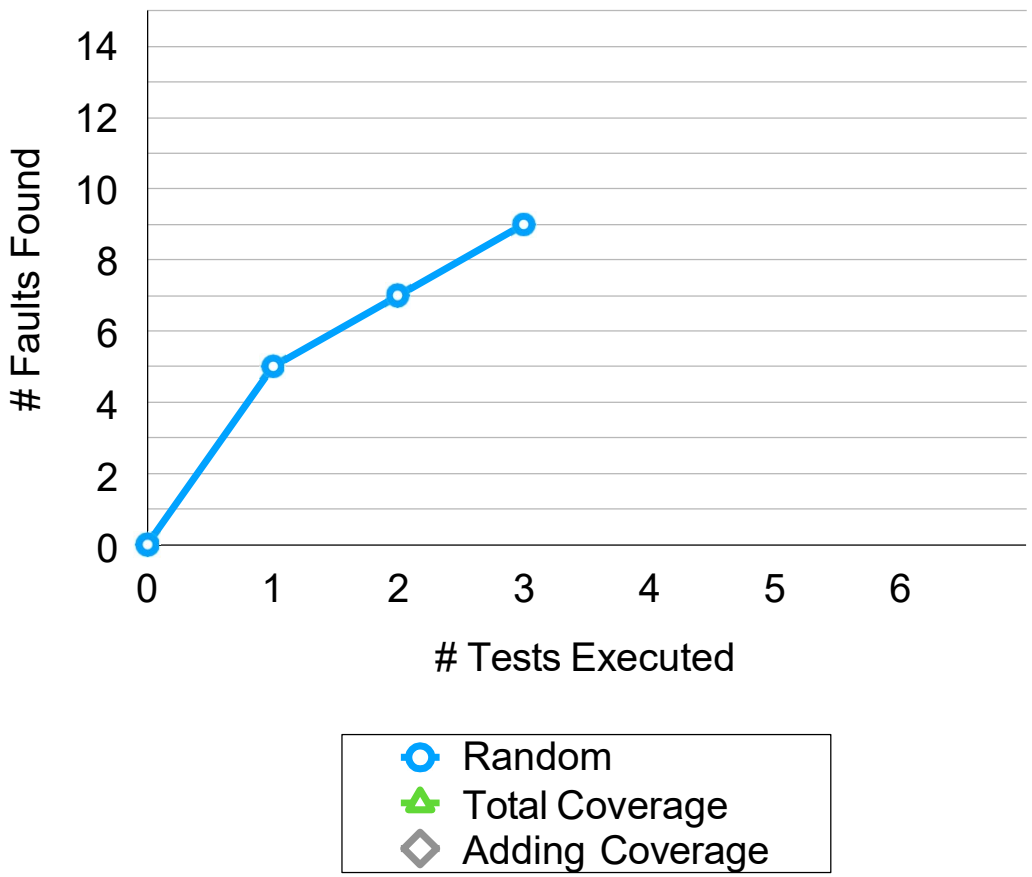
Use Mutation Testing data!

15 mutants in total

Tests A-F kill 13 of them

Random: E-A-B-F-D-C

Using Fault-Detection Data



Use Mutation Testing data!

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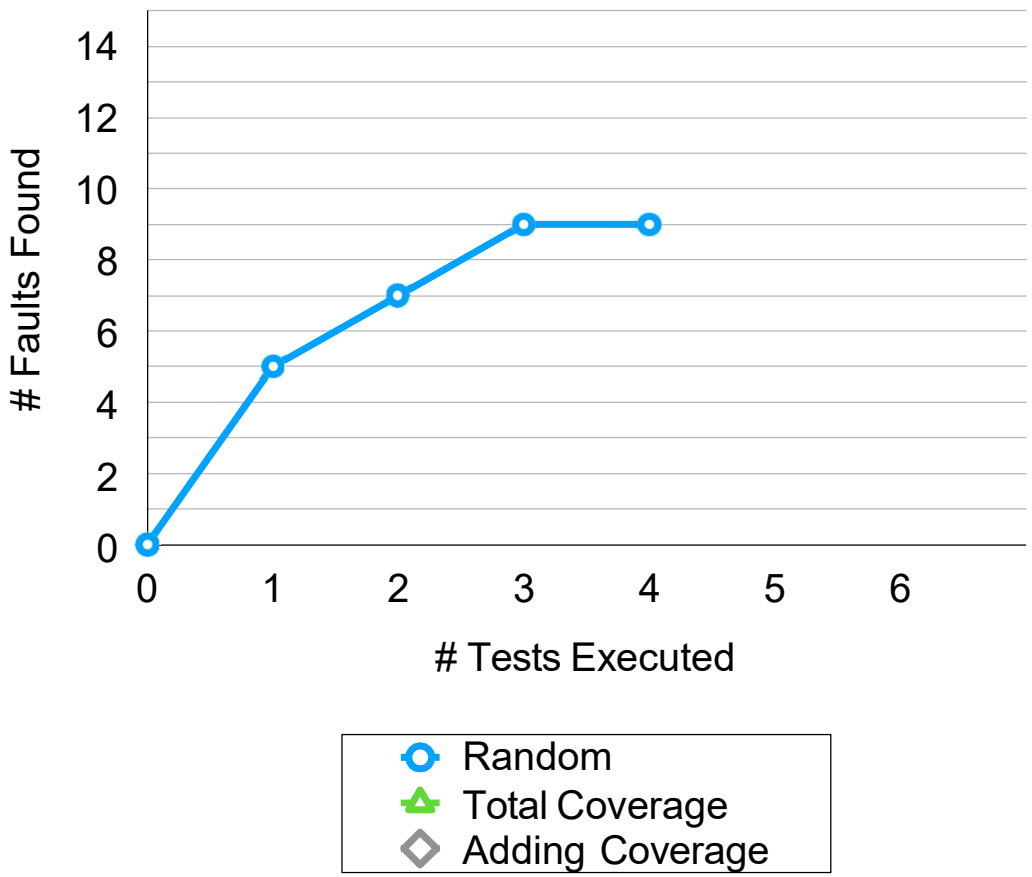
Using Fault-Detection Data



Use Mutation Testing data!

15 mutants in total

Tests A-F kill 13 of them

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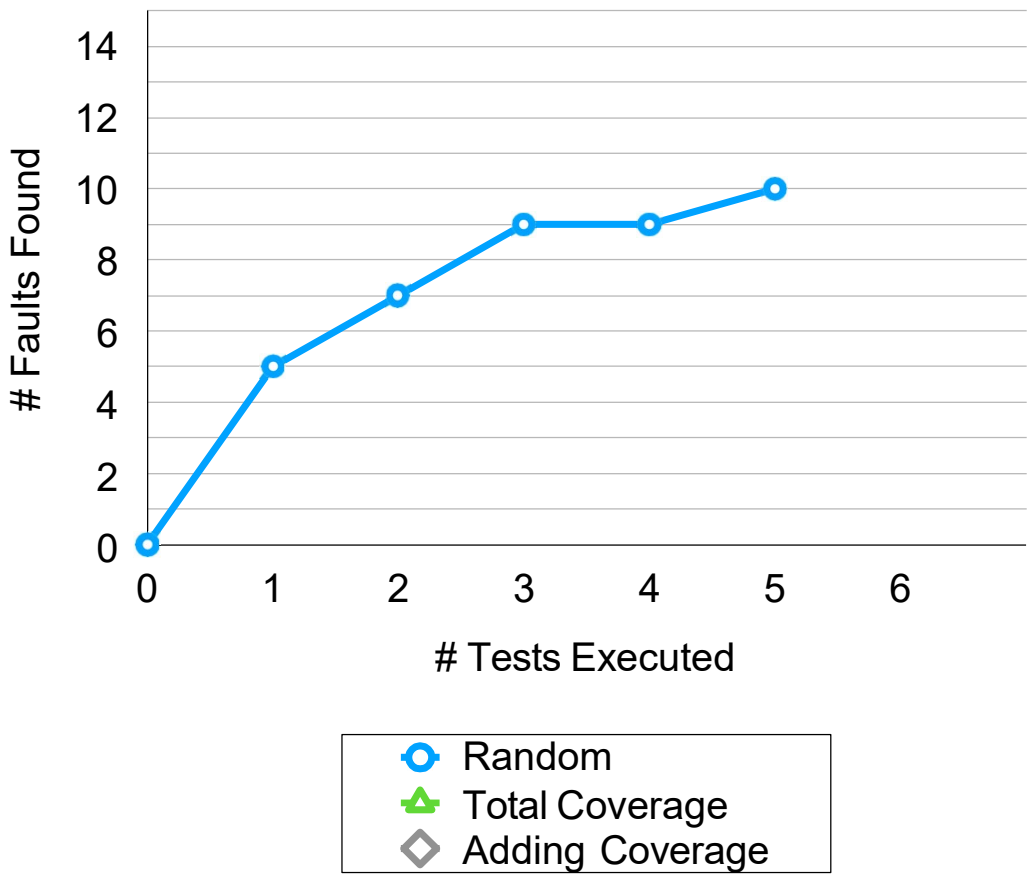
Using Fault-Detection Data



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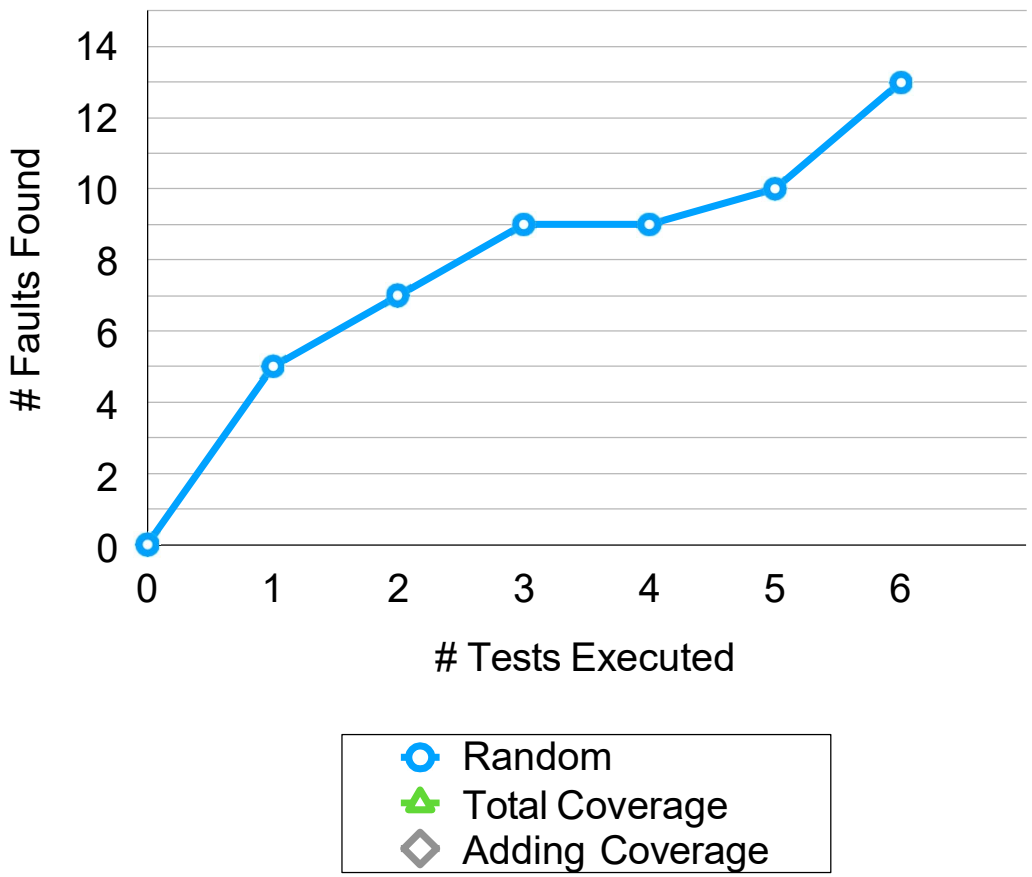

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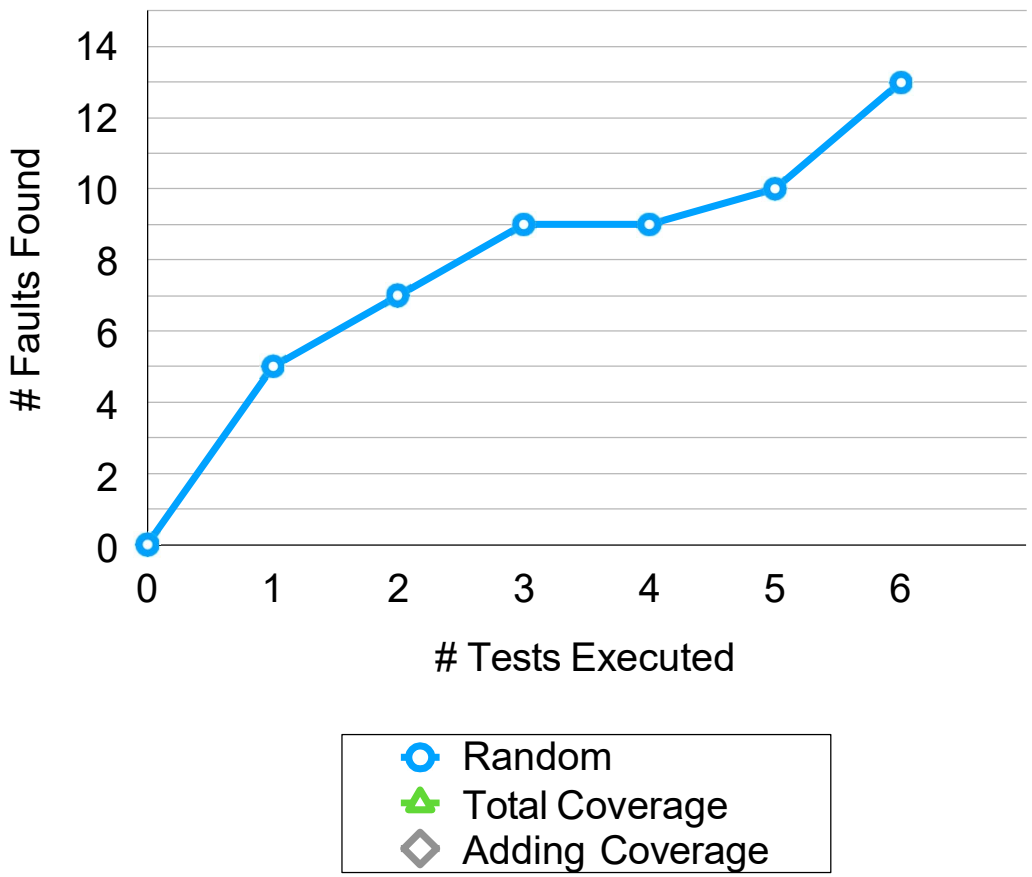

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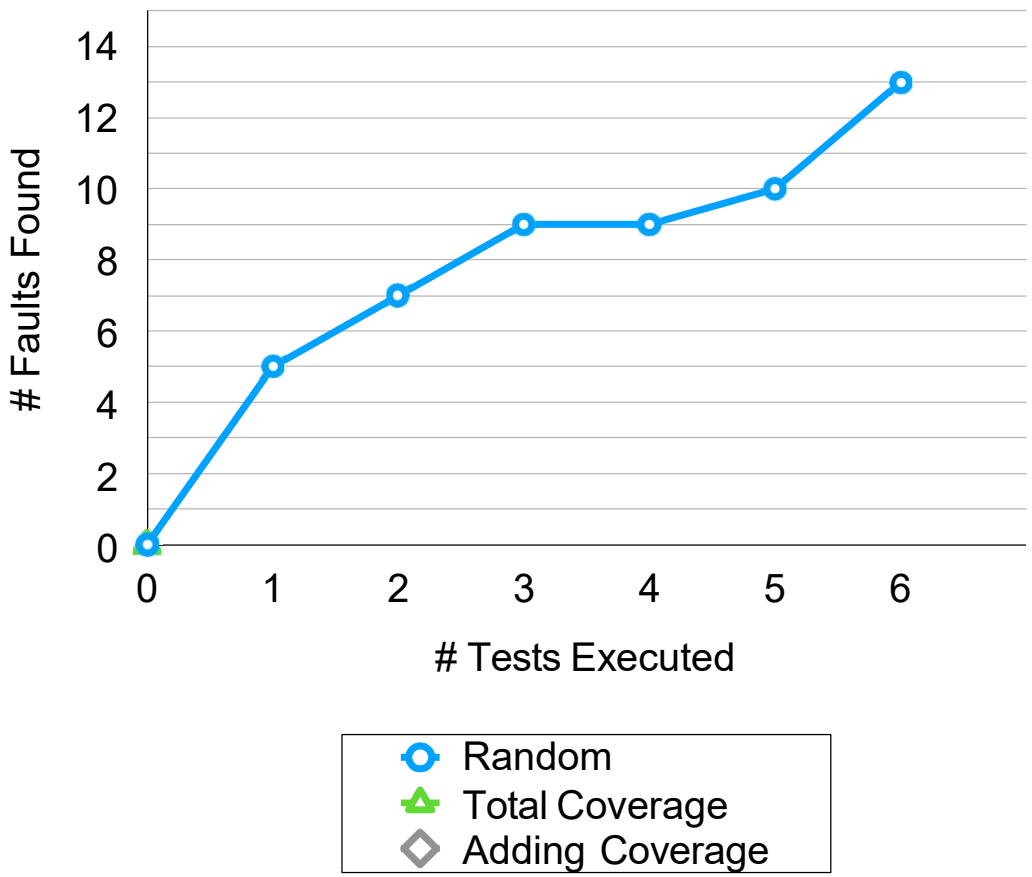

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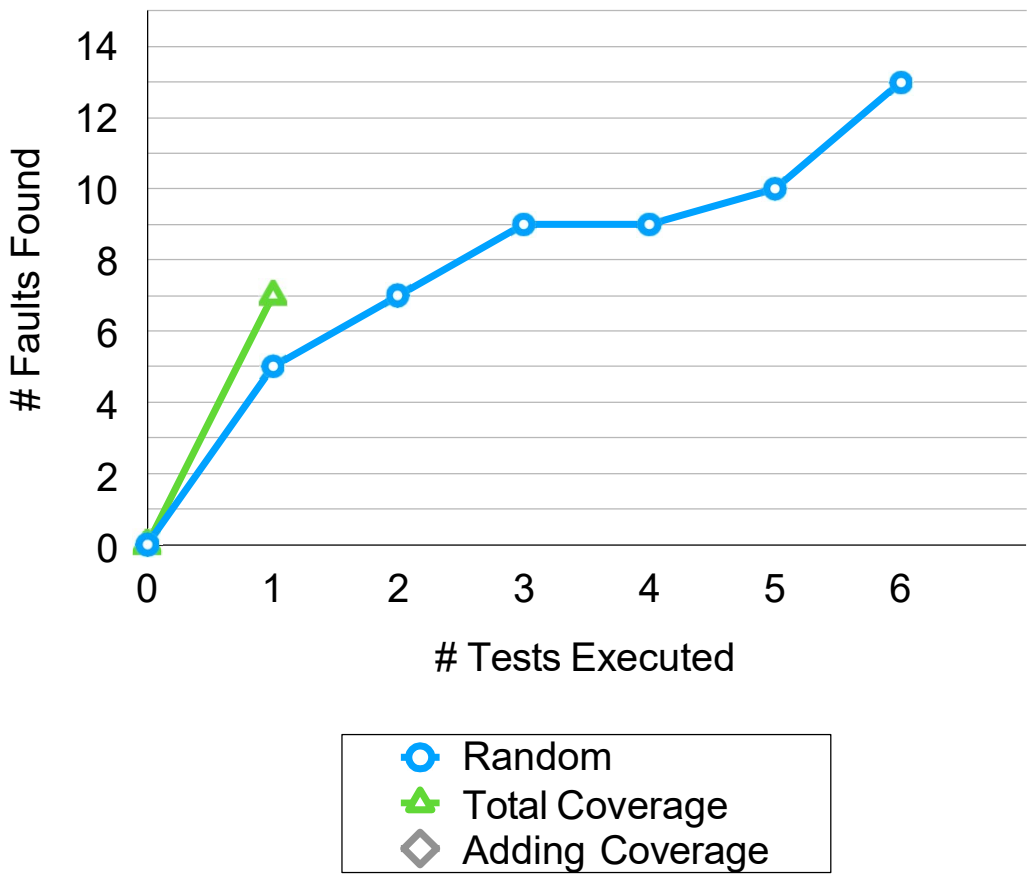

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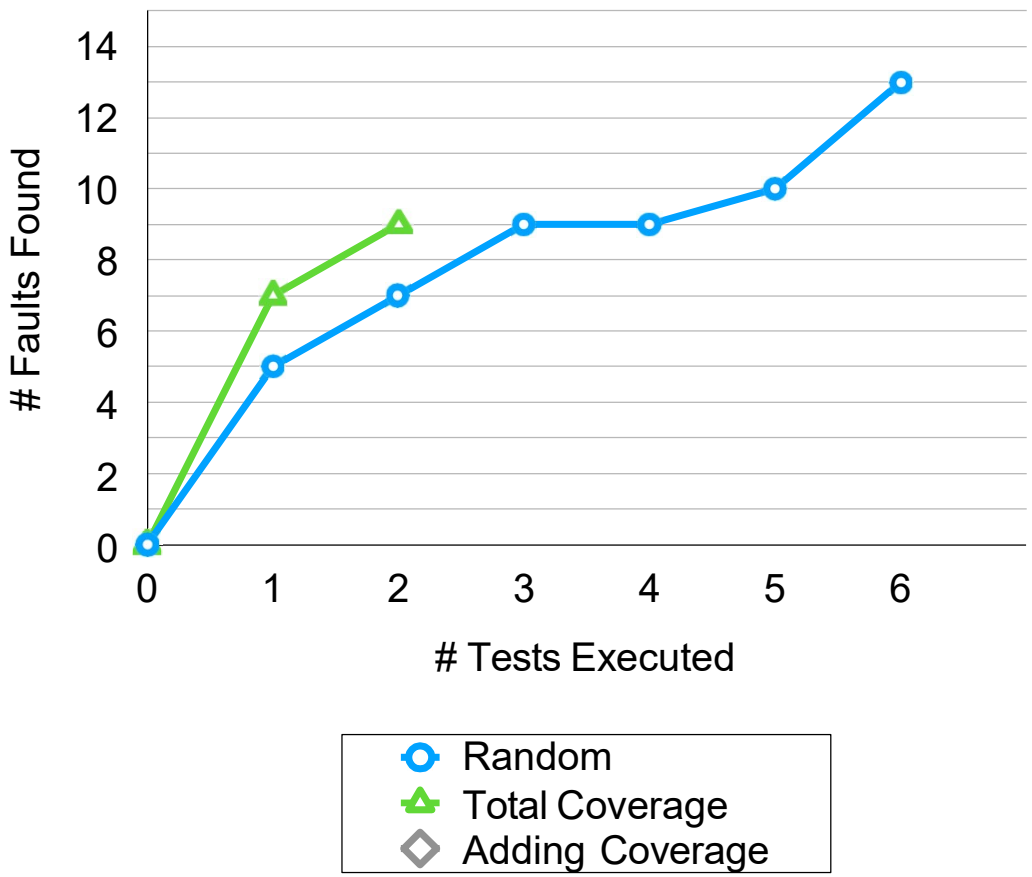

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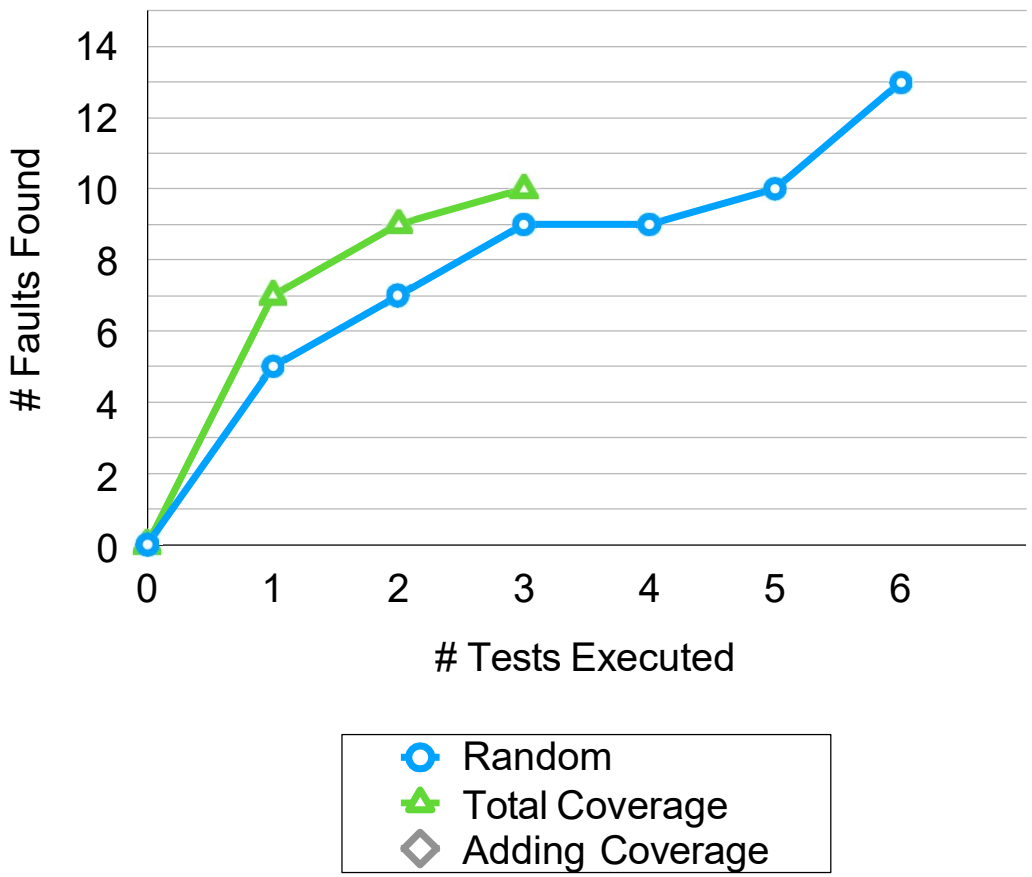

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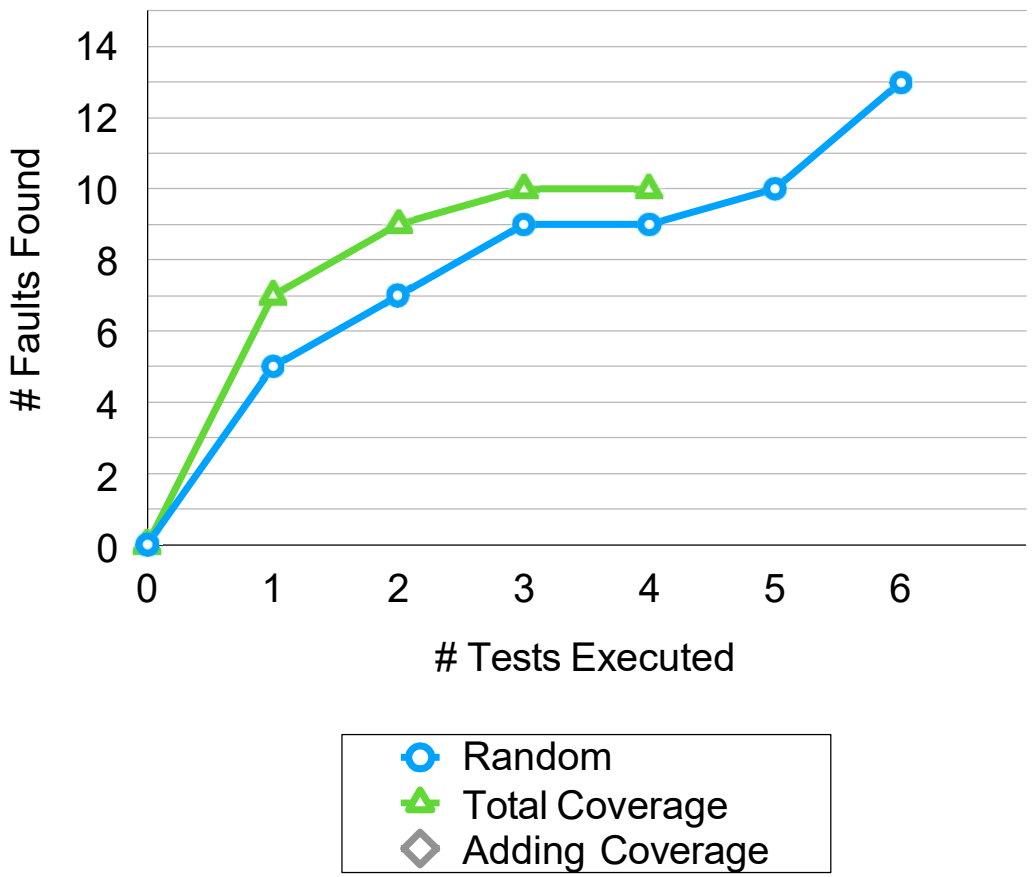

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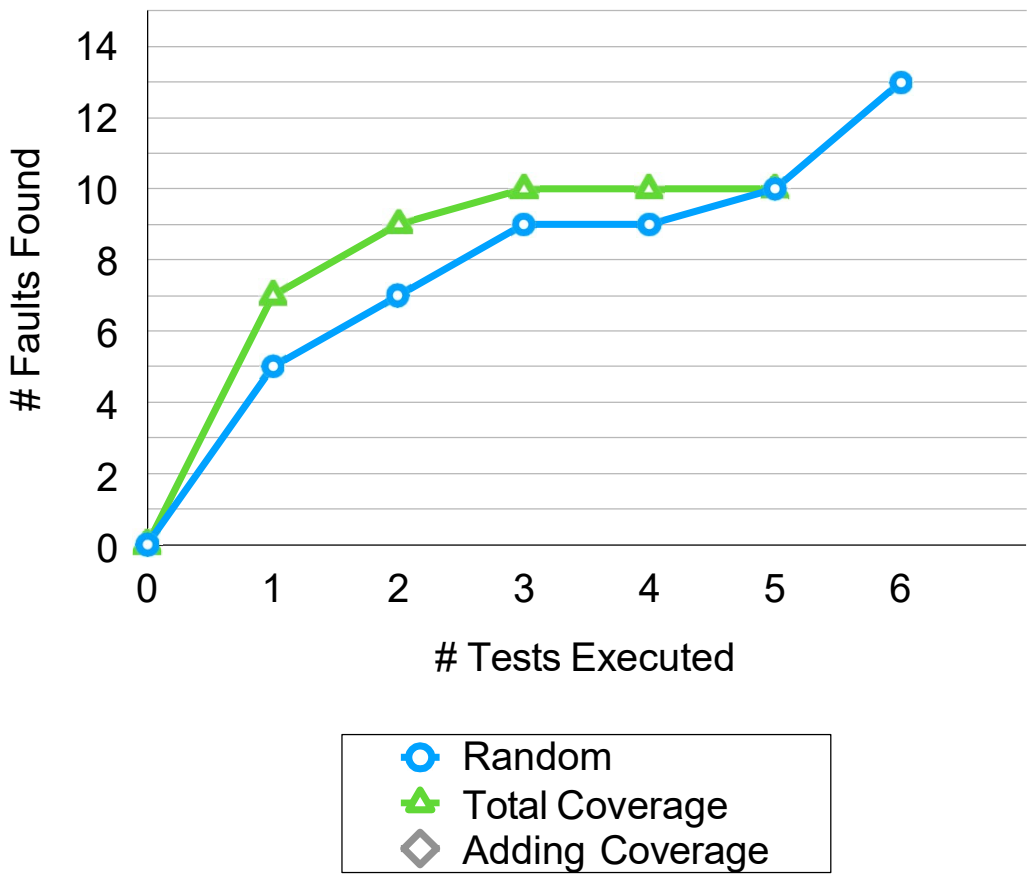

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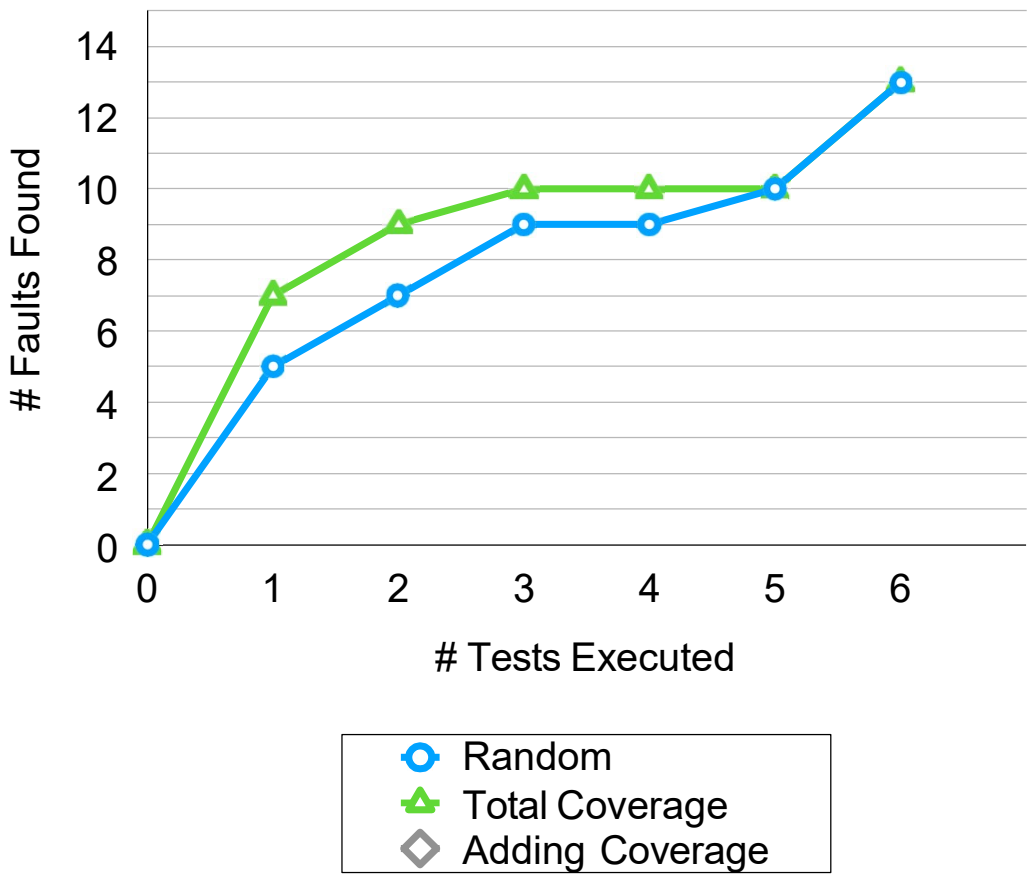

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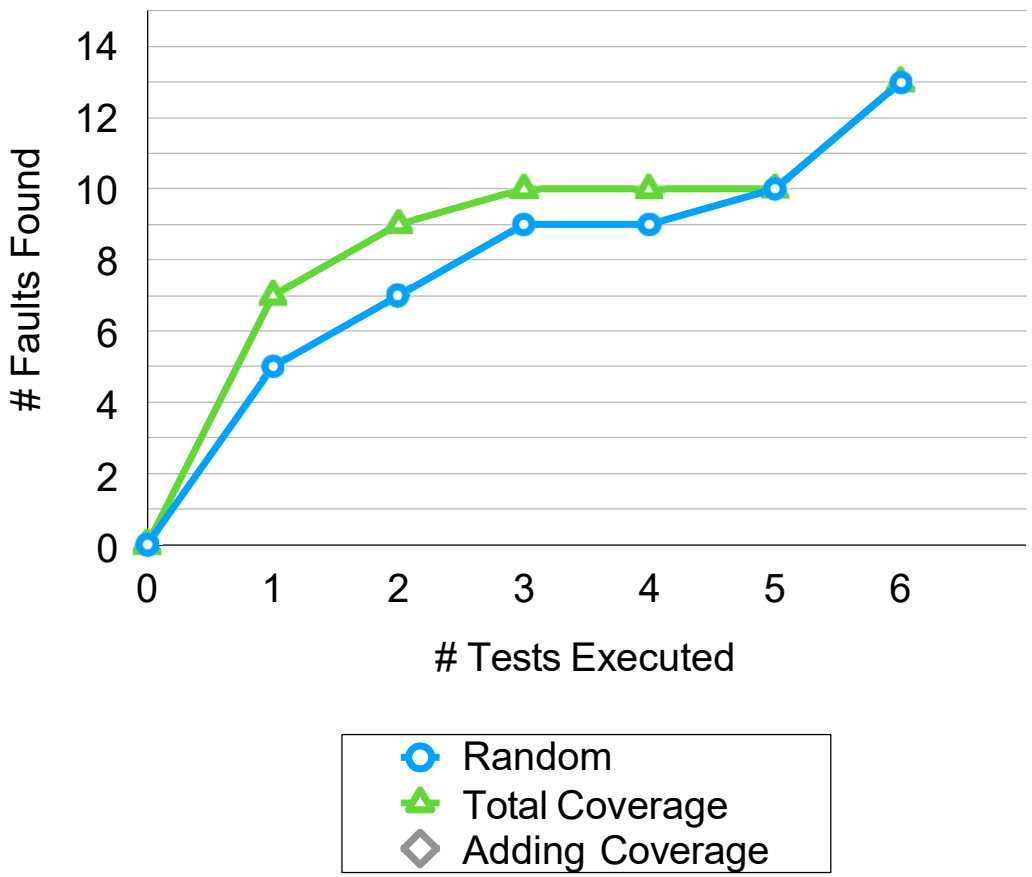

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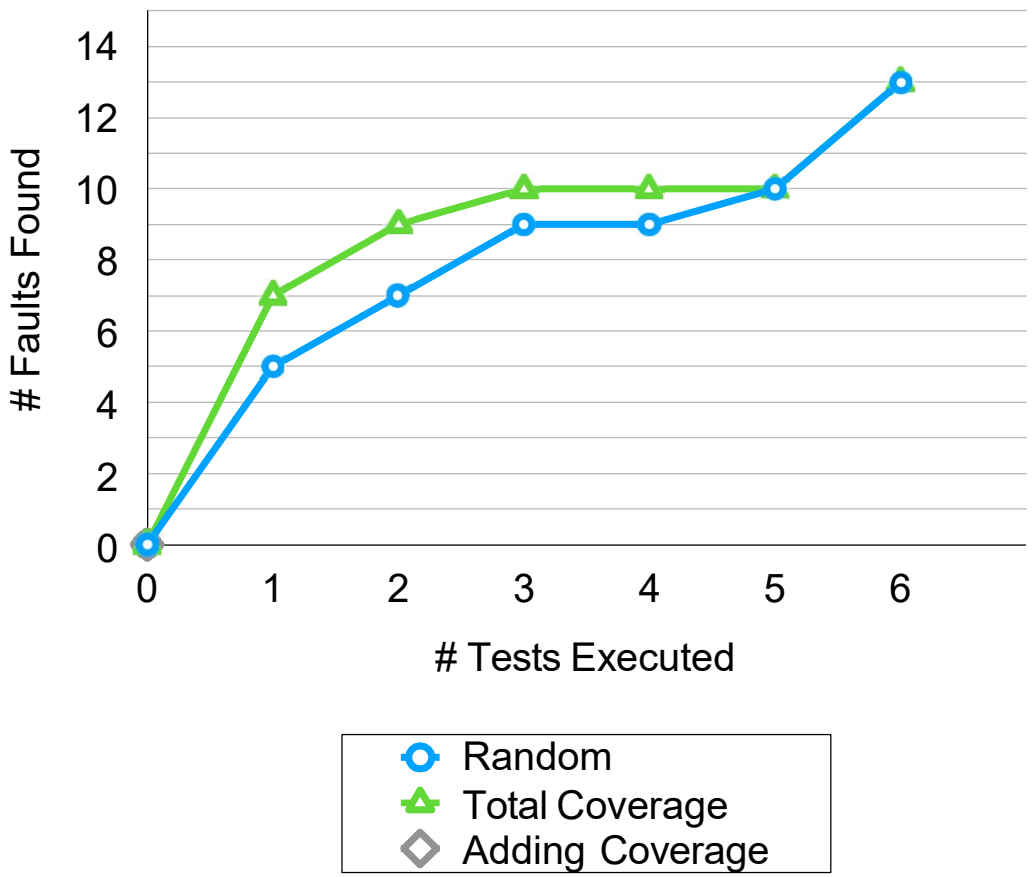

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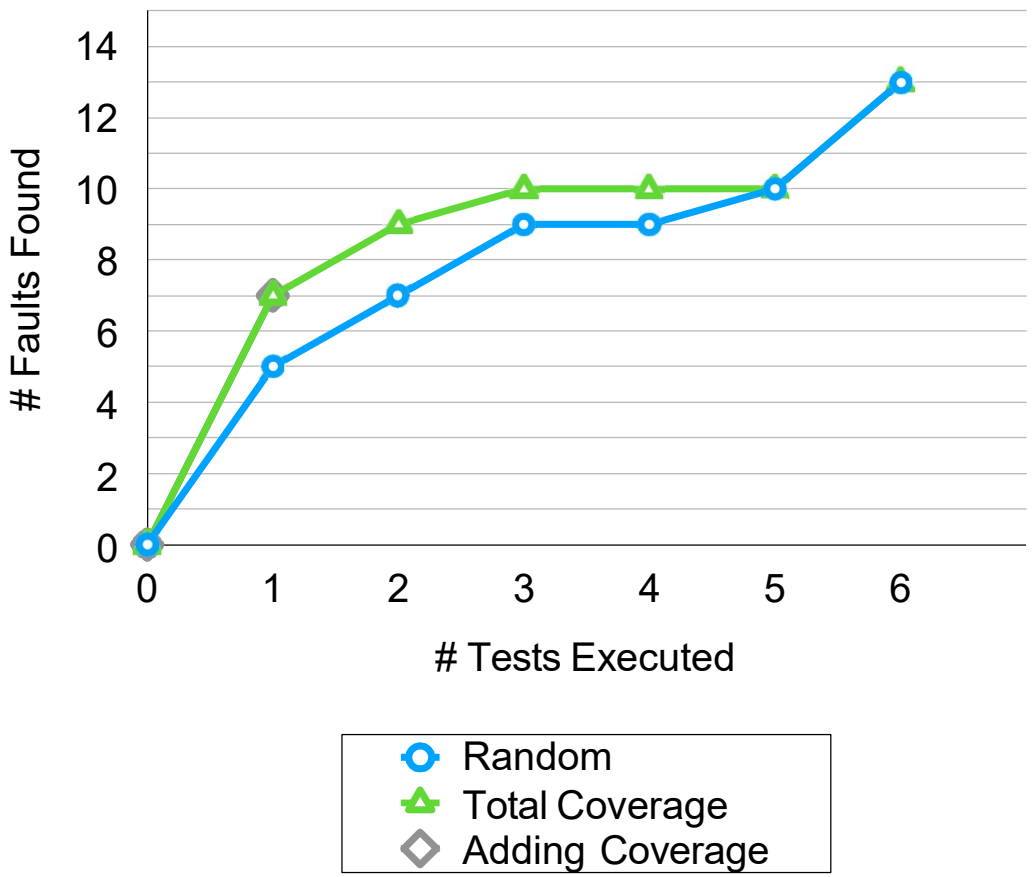

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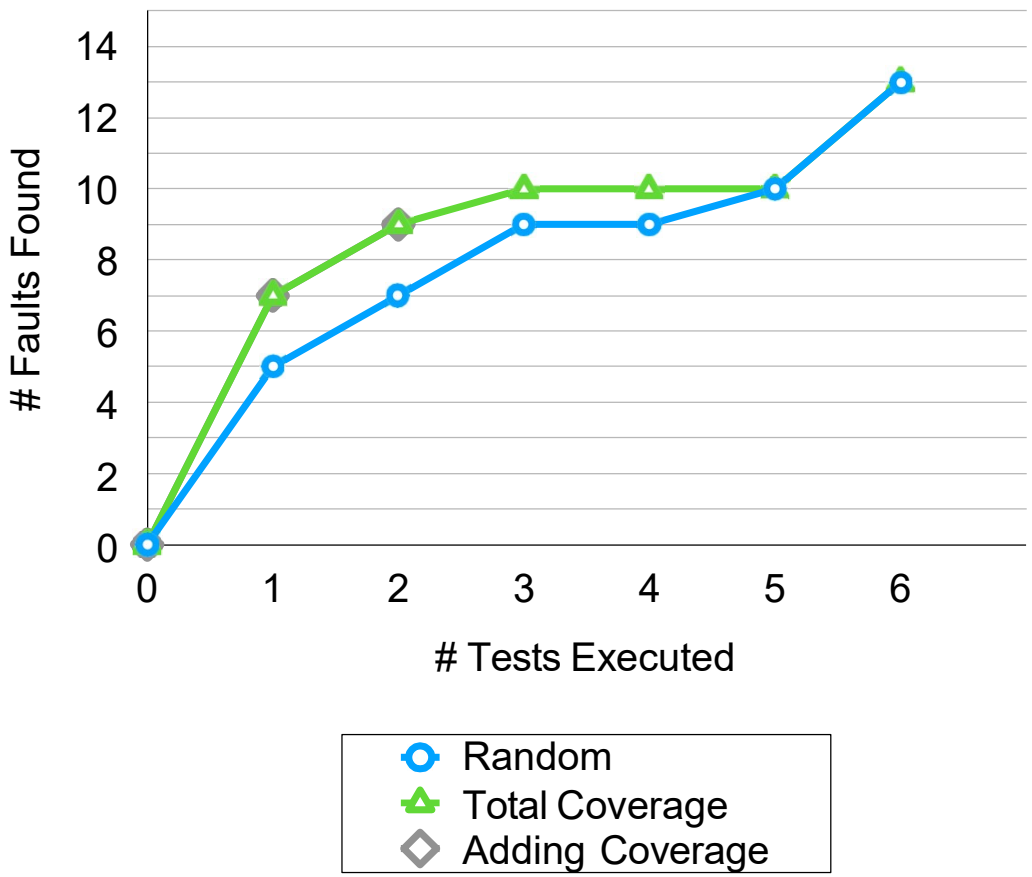

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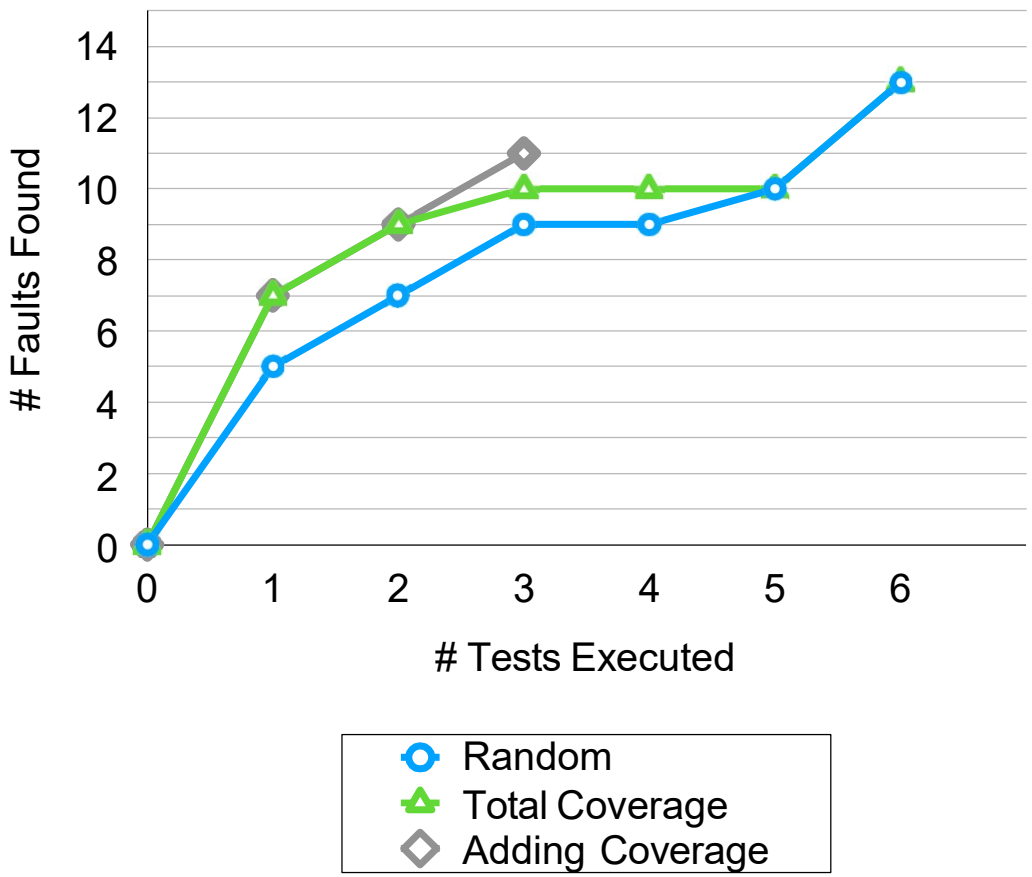

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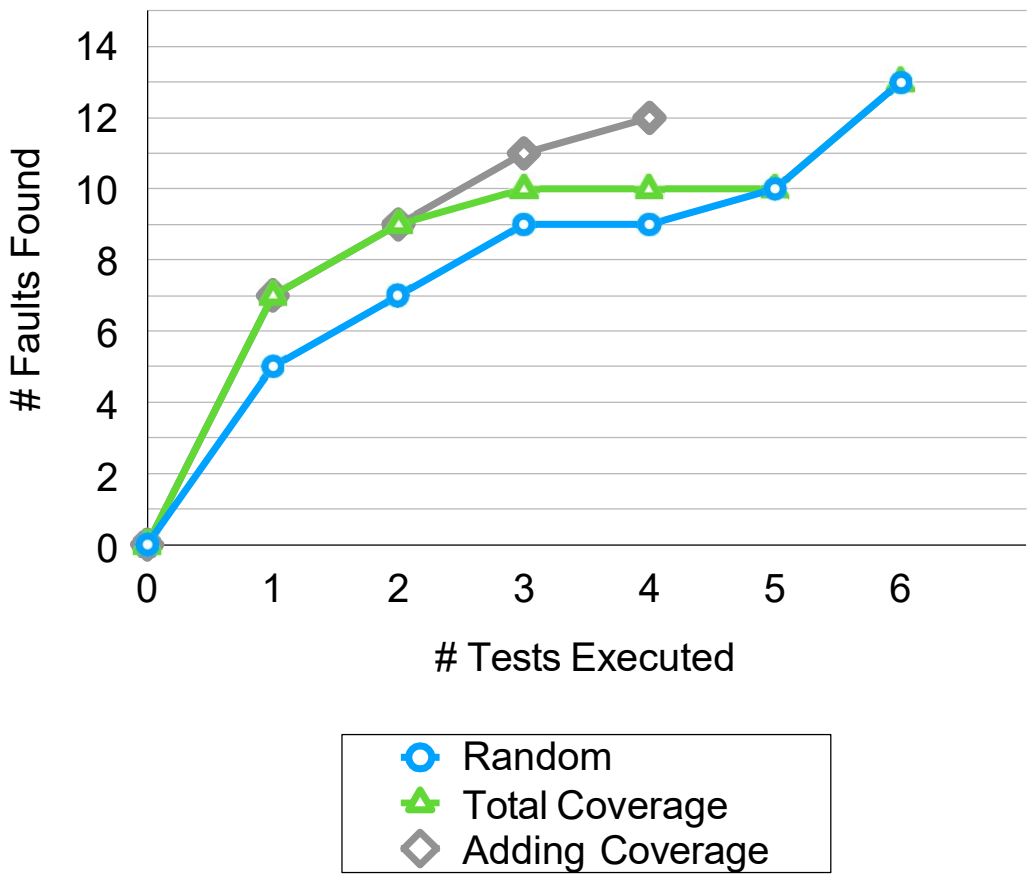

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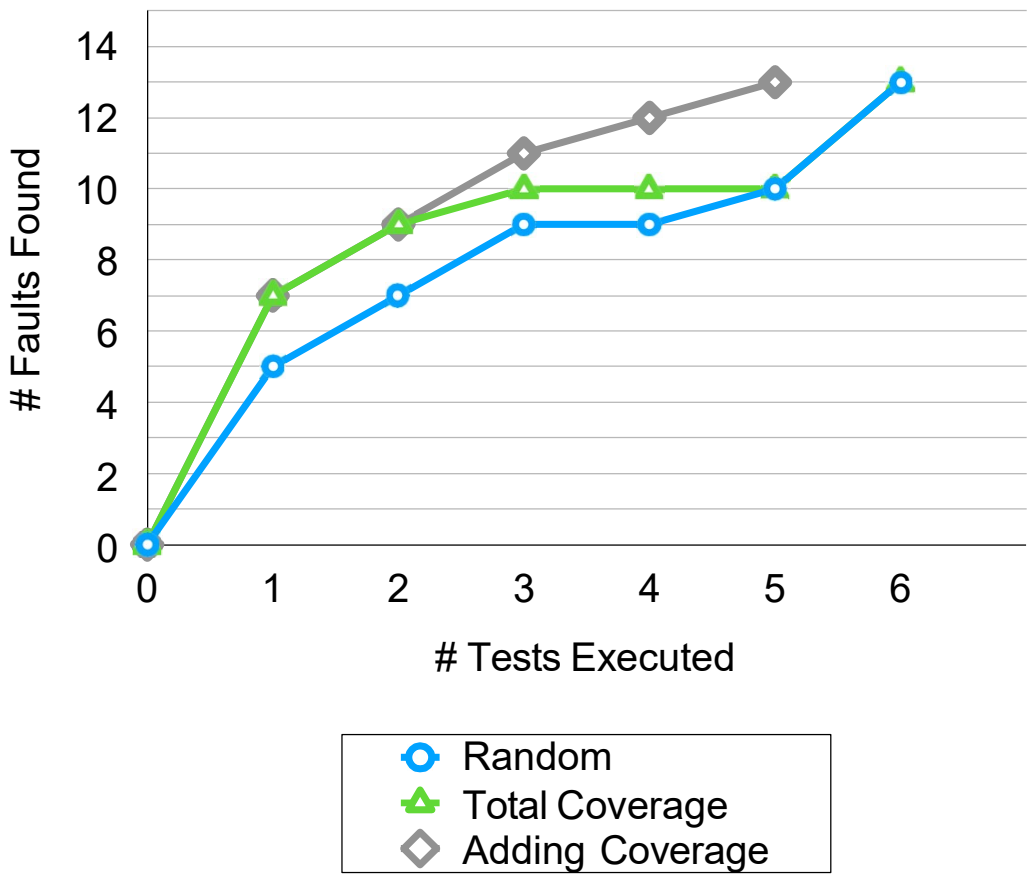

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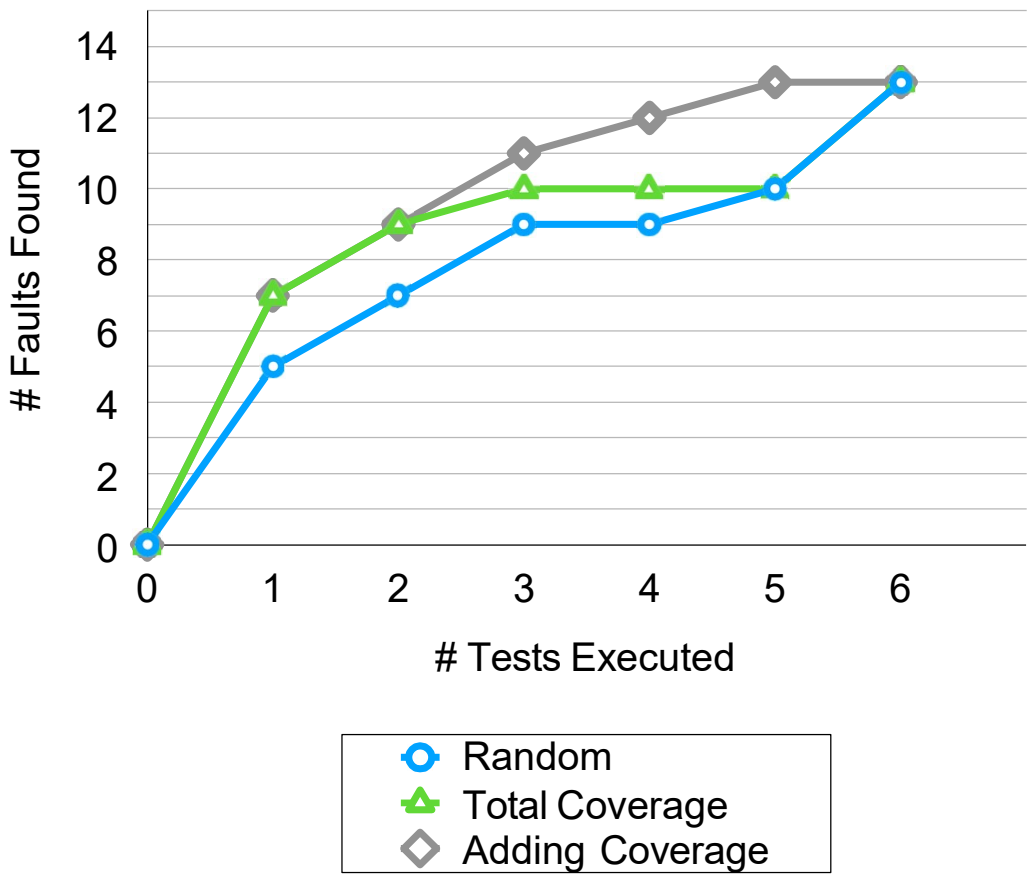

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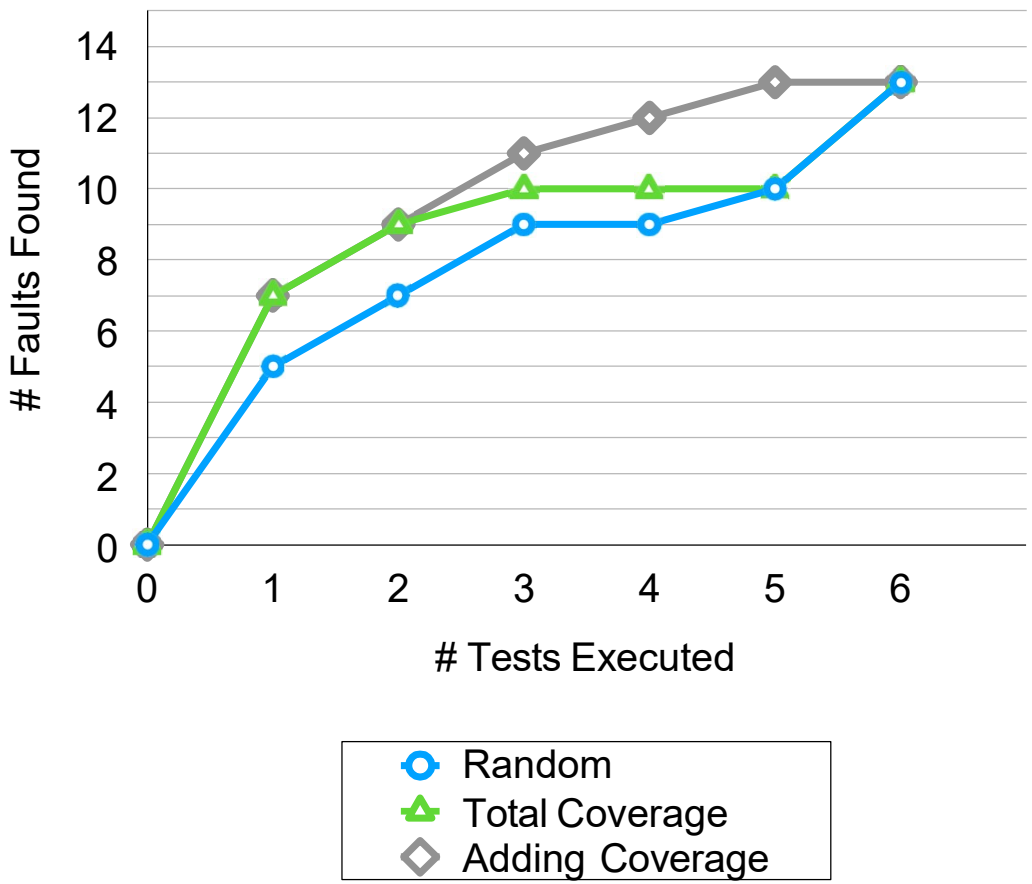

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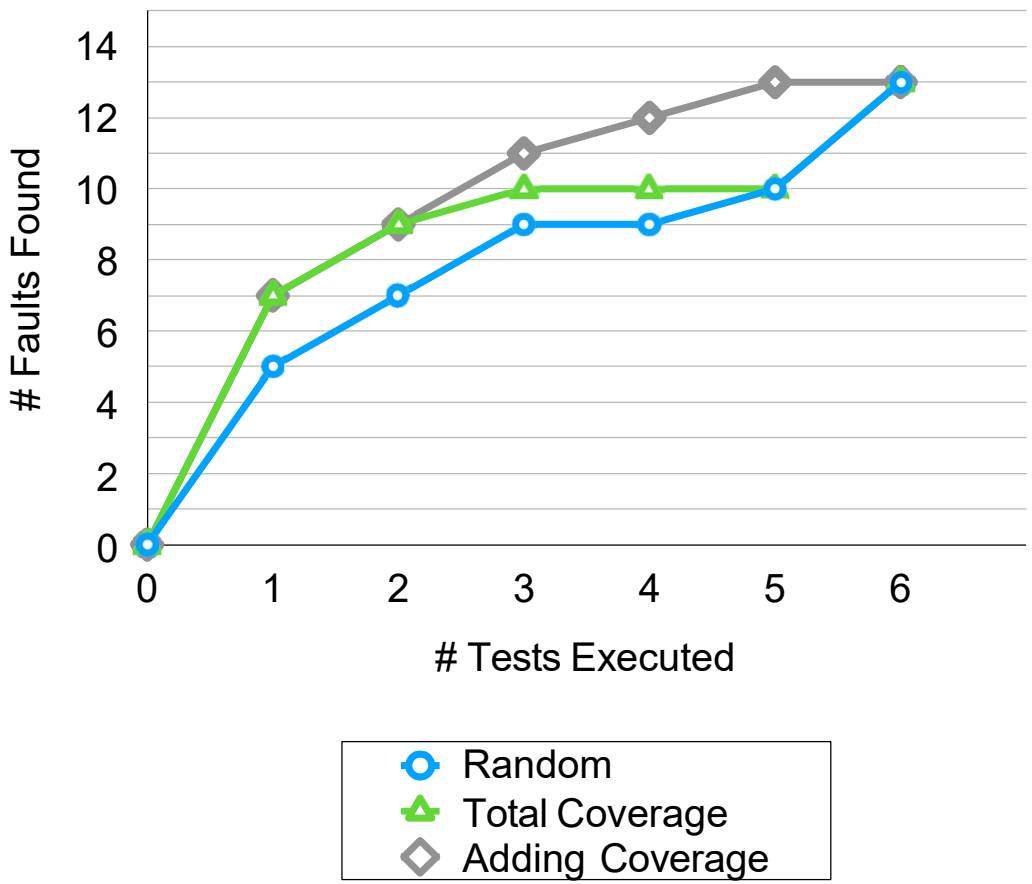

APFD: Average Percentage of Faults Detected

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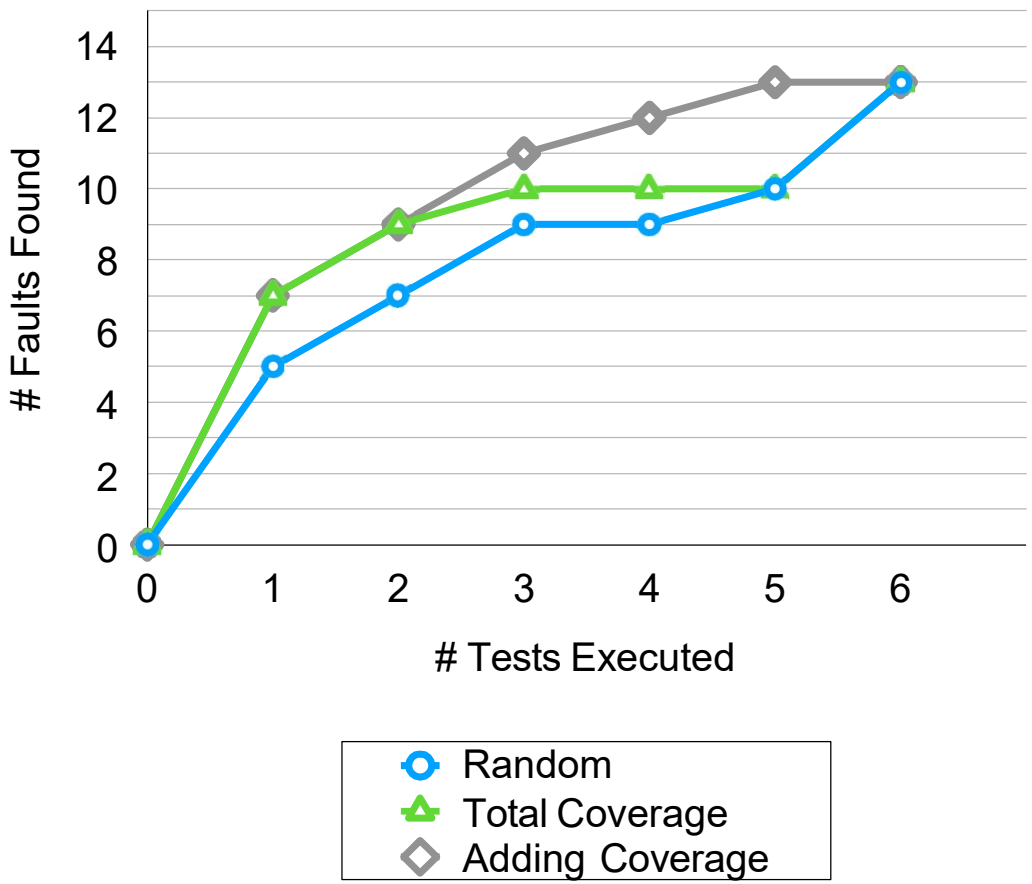
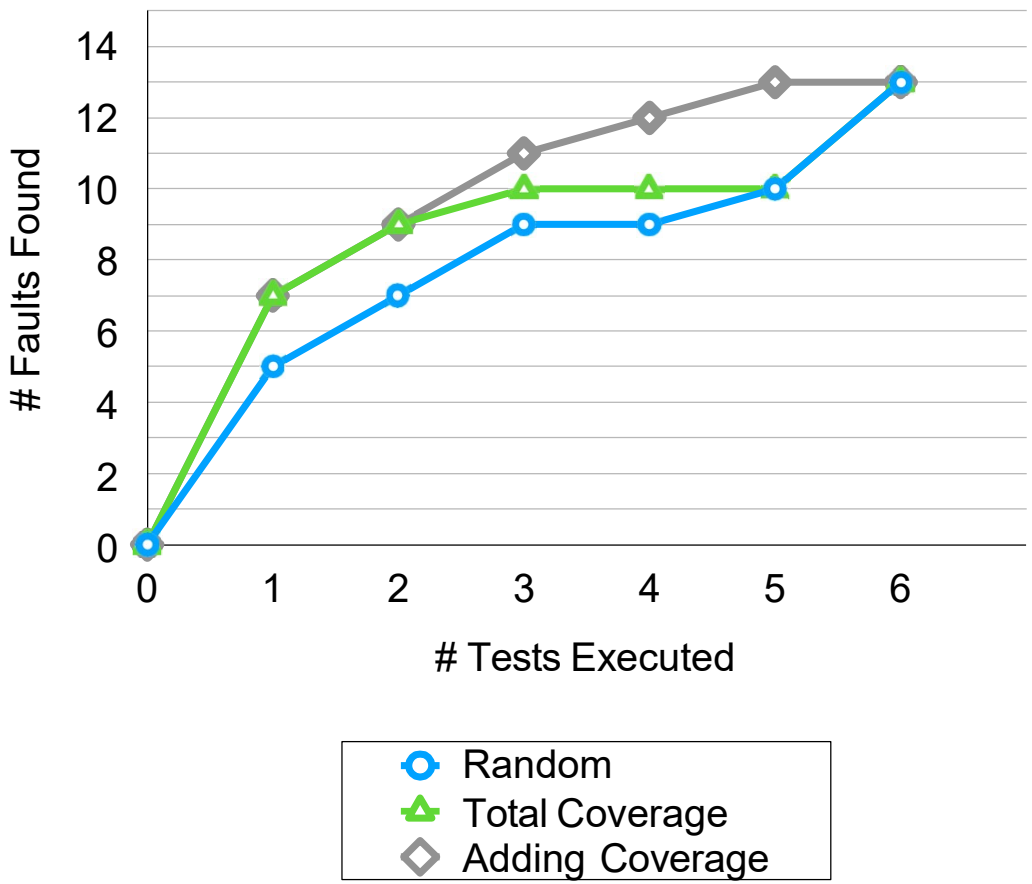


APFD: Average Percentage of Faults Detected

$$APFD = 1 - \frac{TF_1 + TF_2 + \ldots + TF_m}{nm} + \frac{1}{2n}$$

Using Fault-Detection Data



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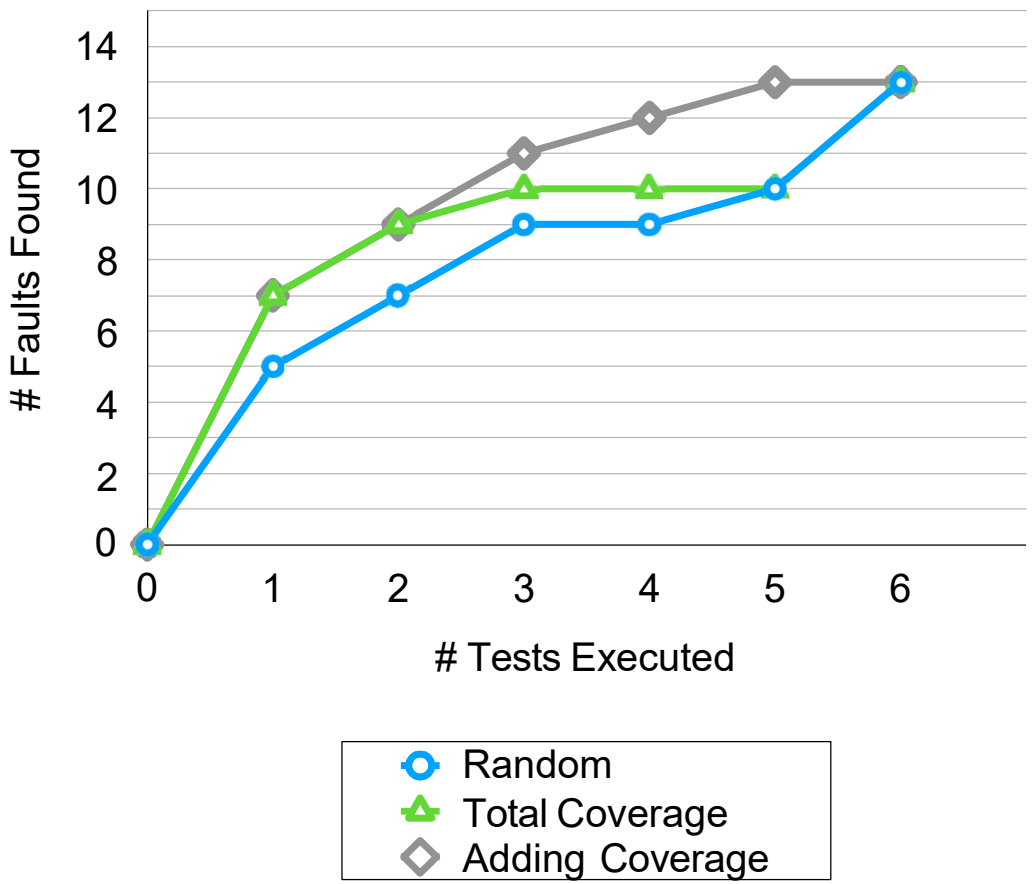

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TF_i is the position of the first test that kills mutant I

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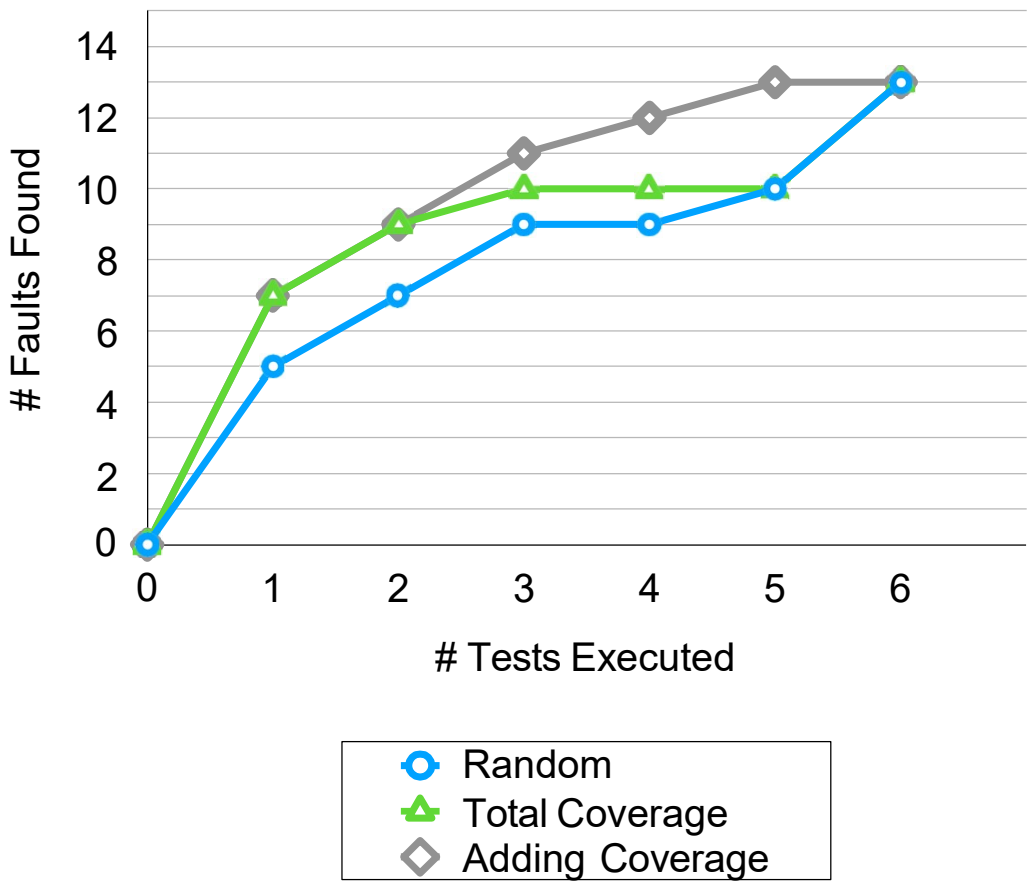

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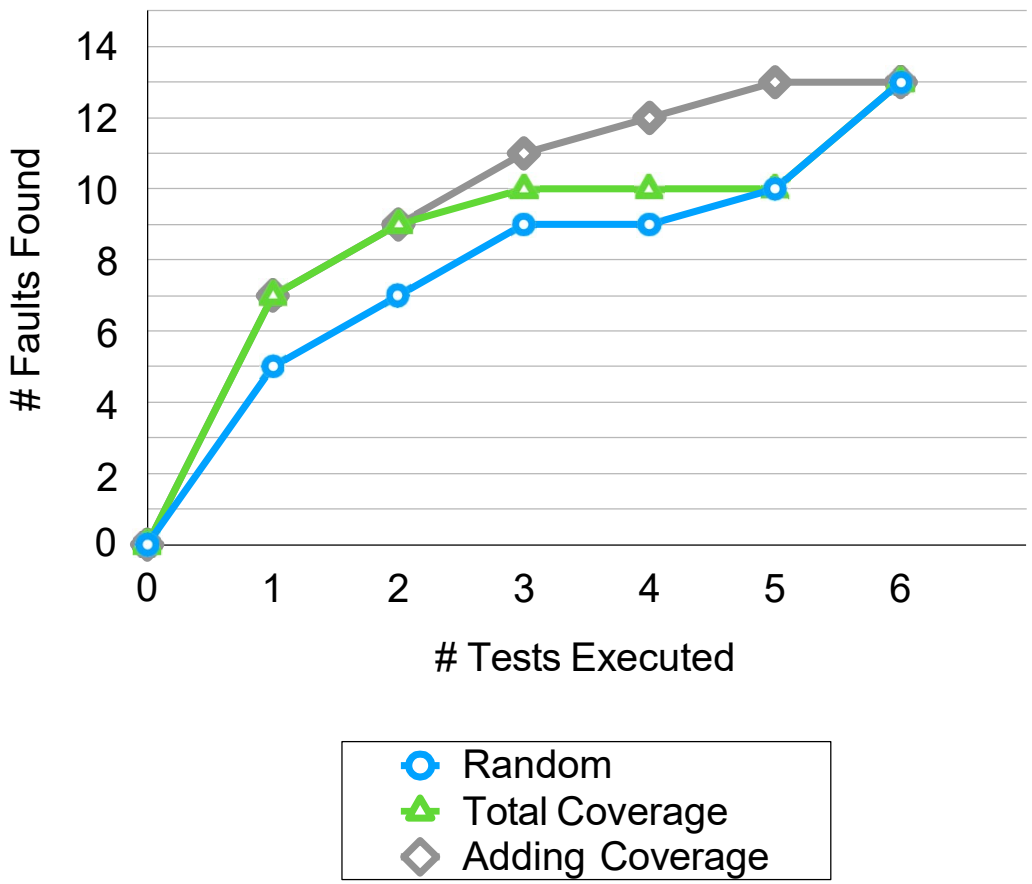

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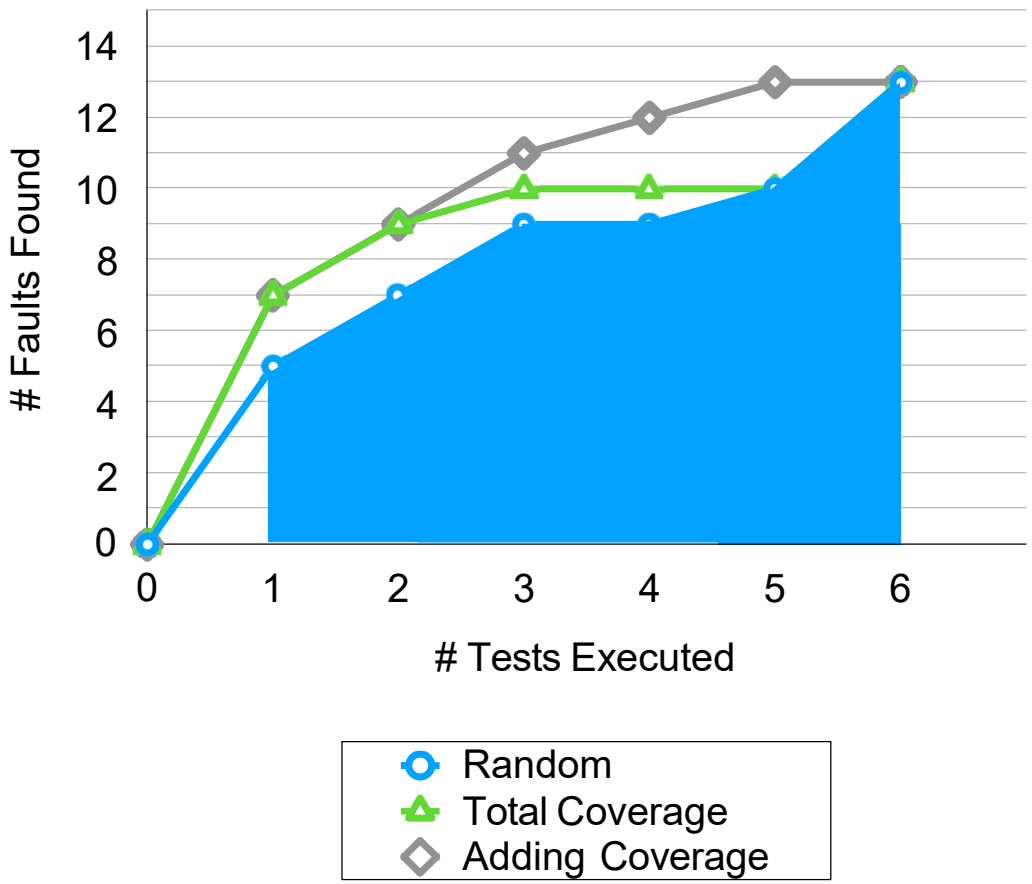

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Fitness Function for Coverage

Consider branch coverage (similar for other metrics)

How to assign a score/fitness to a particular order of tests?

Reward orders that are good for all prefixes

Standard Approach: use weighted average percentage of coverage over the sequence of tests

If we have test cases, branches, and TB_i is the number of the first test case that executes branch , then the fitness is:

$$APC = 1 - \frac{TB_1 + \ldots + TB_m}{nm} + \frac{1}{2n}$$

Optimisation

Similar to minimisation, we might have multiple objectives

- Different forms of coverage.
- Prioritising tests for historically faulty components.
- Prioritising tests for components based on fault prediction techniques.

Potential for Multi-Objective Optimisation

Test Selection

Test Selection

Minimisation aims to remove redundant tests

- **But is this the best idea in the long term?**
- A removed test may have been **useful to reveal a bug in the future**

Selection is an alternative to **minimisation**

- It's ok to keep large test suites in the codebase
- As long as which tests are run upon every change is decided carefully

Identification of affected code + Selection of tests to run

Intra-Procedural Test Selection

Identify **affected** code based on **parallel traversal** of the CFGs (control flow graphs) of the original and the modified versions

At each step: compare nodes for **lexicographical equivalence**

Difference found? **Select** tests traversing path from entry node to changed node, i.e., **modification-traversing tests**

An Average Example

```
public static int avg(File inputFile) throws Exception {
 1  int count = 0; int sum = 0;
 2  Scanner in = new Scanner(new FileReader(inputFile));
 3  while(in.hasNext()) {
 4      String line = in.next();
 5      if (! StringUtils.isNumeric(line)) {
 6          throw new NumberFormatException();
 7      } else {
 8          sum += Integer.parseInt(line);
 9          count++;
10      }
11  }
12  in.close();
13  return count > 0 ? sum / count : 0;
}
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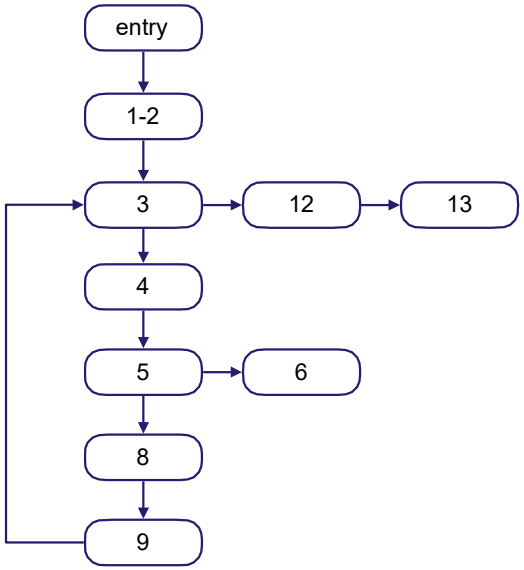

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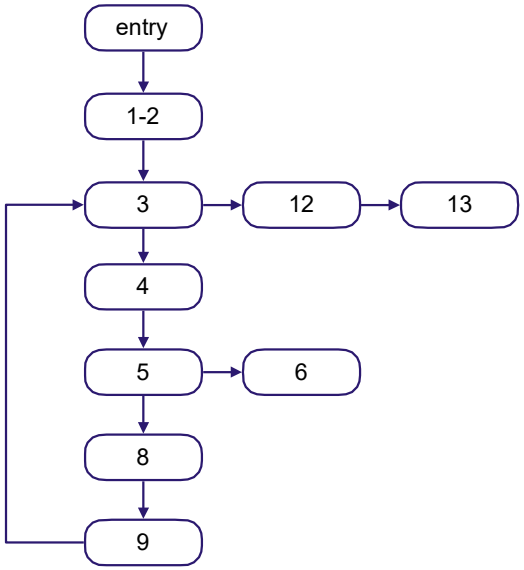


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@Test
public void t1() throws Exception {
    assertEquals(0, TestSelection.avg(empty));
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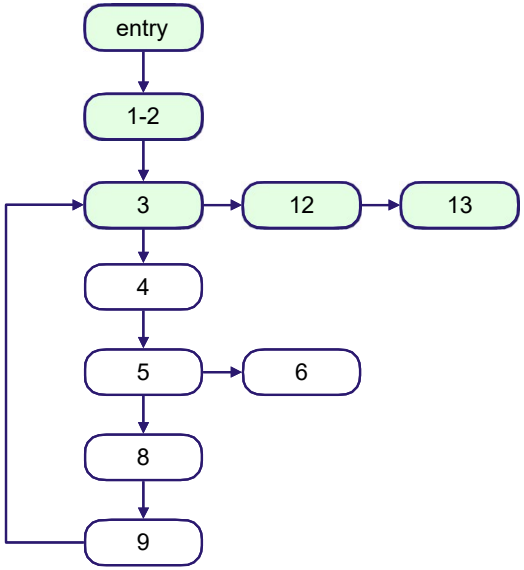


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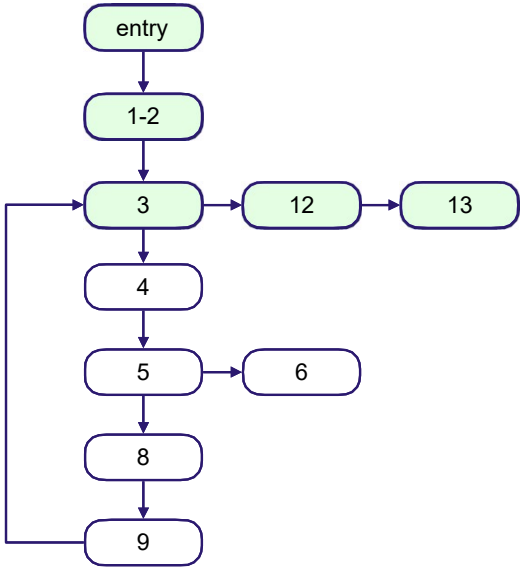


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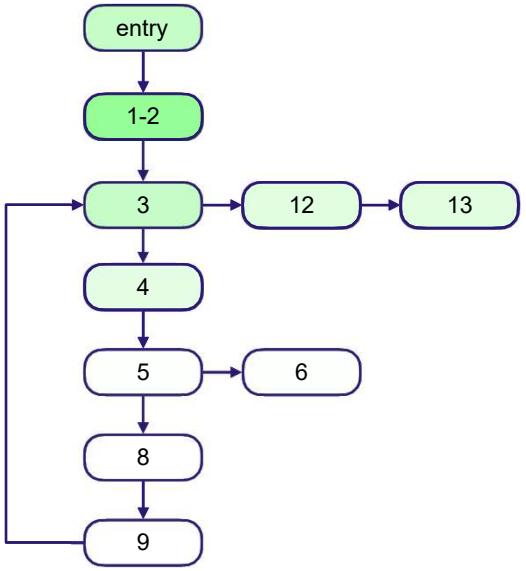


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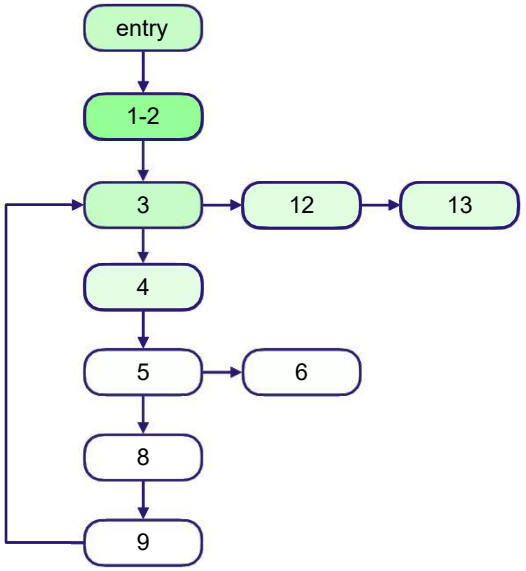


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@Test
public void t3() throws Exception {
    assertEquals(2, TestSelection.avg(numbers123));
}
```

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 1  int count = 0; int sum = 0;
 2  Scanner in = new Scanner(new FileReader(inputFile));
 3  while(in.hasNext()) {
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 5      if (! StringUtils.isNumeric(line)) {
 6          throw new NumberFormatException();
 7      } else {
 8          sum += Integer.parseInt(line);
 9          count++;
10      }
11  }
12  in.close();
13  return count > 0 ? sum / count : 0;
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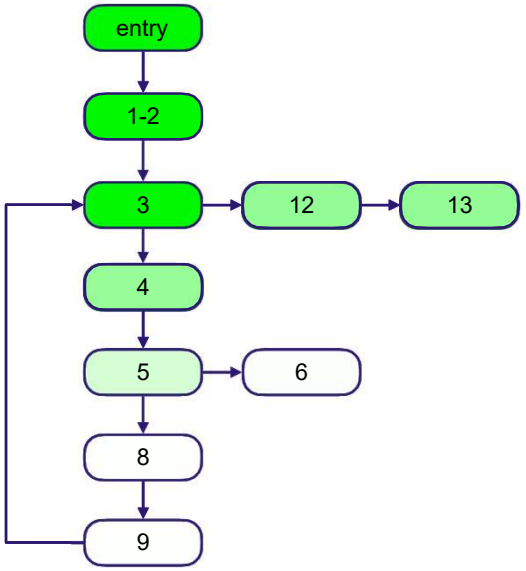
```

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@Test
public void t1() throws Exception {
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}

@Test
public void t2() throws Exception {
    assertThrows(NumberFormatException.class, () -> {
        TestSelection.avg(nonNumeric);
    });
}

@Test
public void t3() throws Exception {
    assertEquals(2, TestSelection.avg(numbers123));
}
```

An Average Example

```
public static int avg(File inputFile) throws Exception {
 1  int count = 0; int sum = 0;
 2  Scanner in = new Scanner(new FileReader(inputFile));
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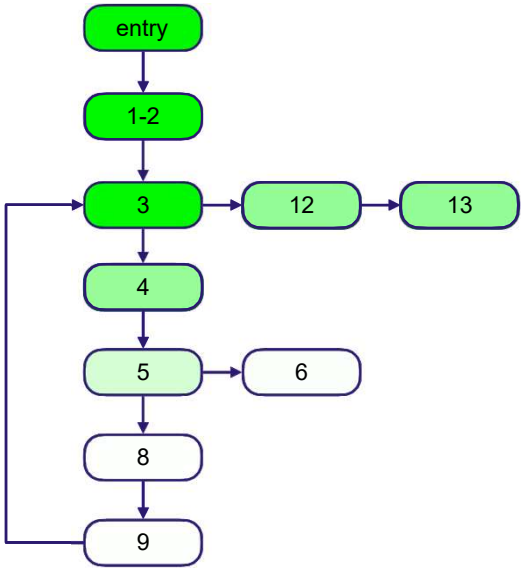


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Test	Edges Traversed

An Average Example

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public static int avg(File inputFile) throws Exception {
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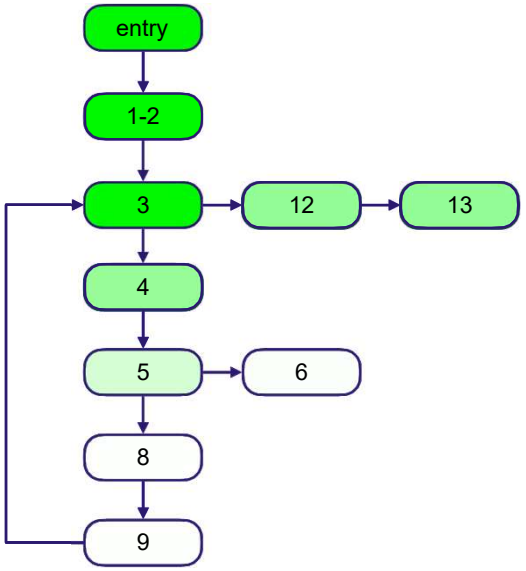


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public void t3() throws Exception {
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```

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)

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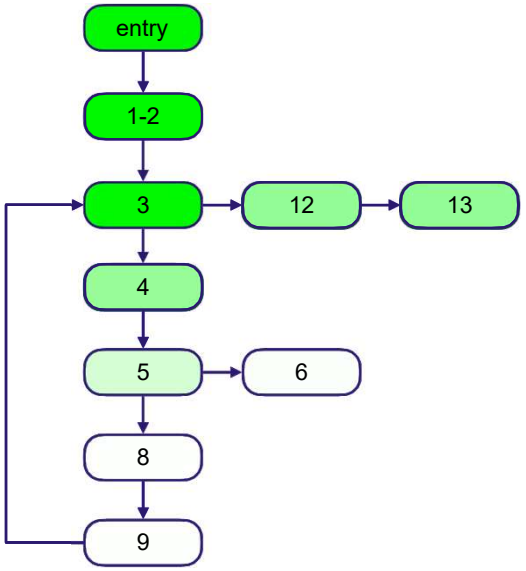


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public void t3() throws Exception {
    assertEquals(2, TestSelection.avg(numbers123));
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```

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
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}
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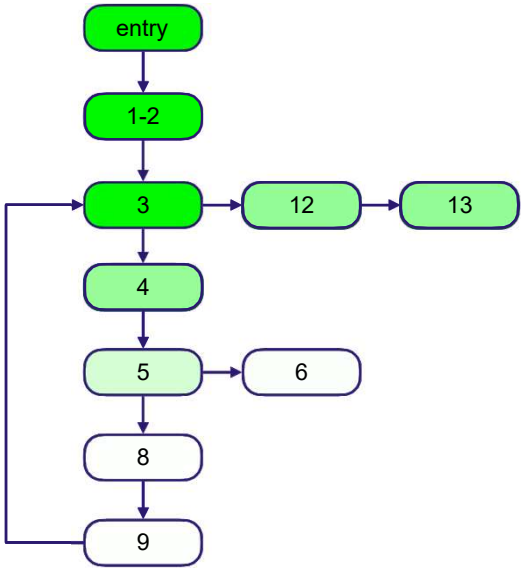


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 4      String line = in.next();
 5      if (! StringUtils.isNumeric(line)) {
 6          throw new Exception();
 7      } else {
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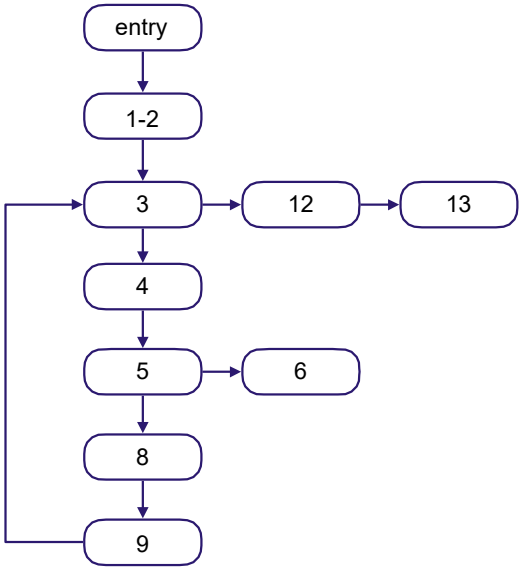


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public static int avg(File inputFile) throws Exception {
 1  int count = 0; int sum = 0;
 2  Scanner in = new Scanner(new FileReader(inputFile));
 3  while(in.hasNext()) {
 4      String line = in.next();
 5      if (! StringUtils.isNumeric(line)) {
 6          System.err.println("Bad input");
 7          throw new Exception();
 8      } else {
 9          sum += Integer.parseInt(line);
10      }
11  }
12  in.close();
13  return count > 0 ? sum / count : 0;
}
```
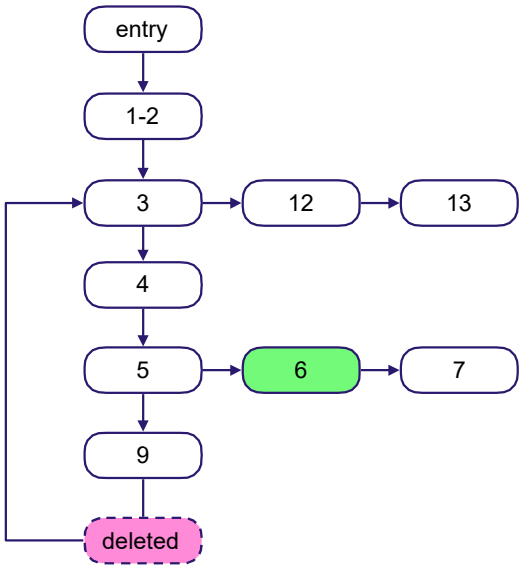
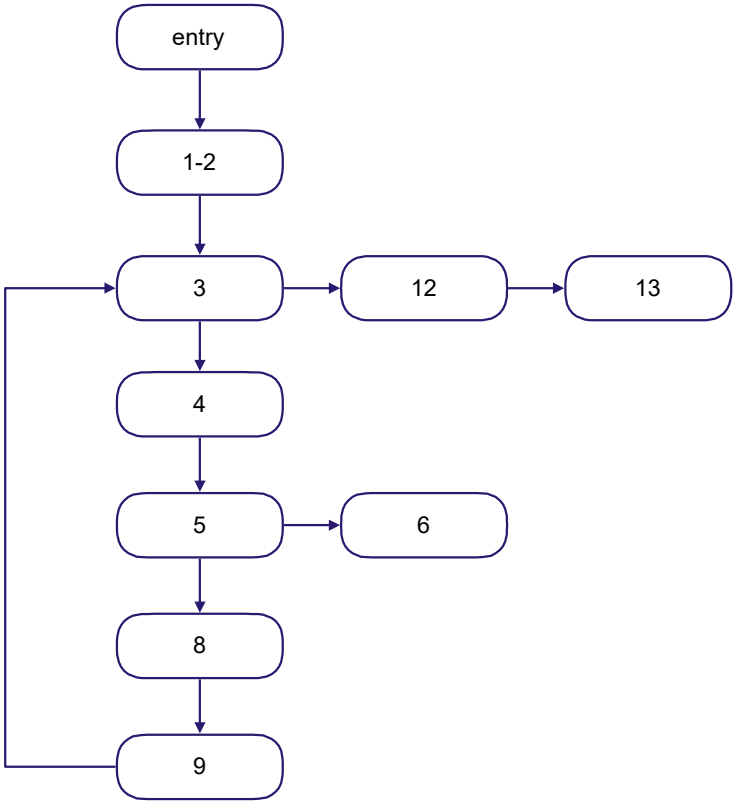
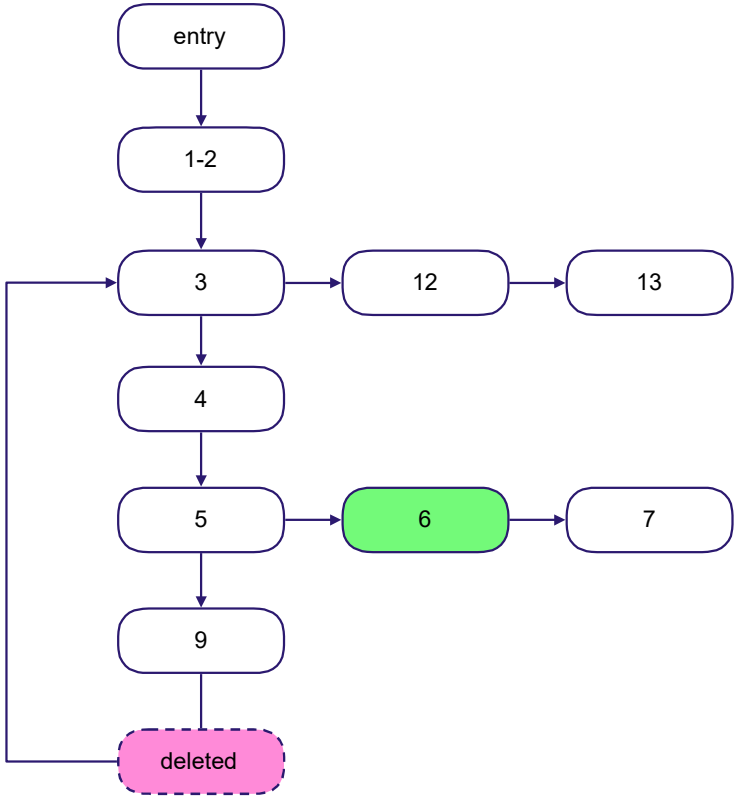



An Average Example

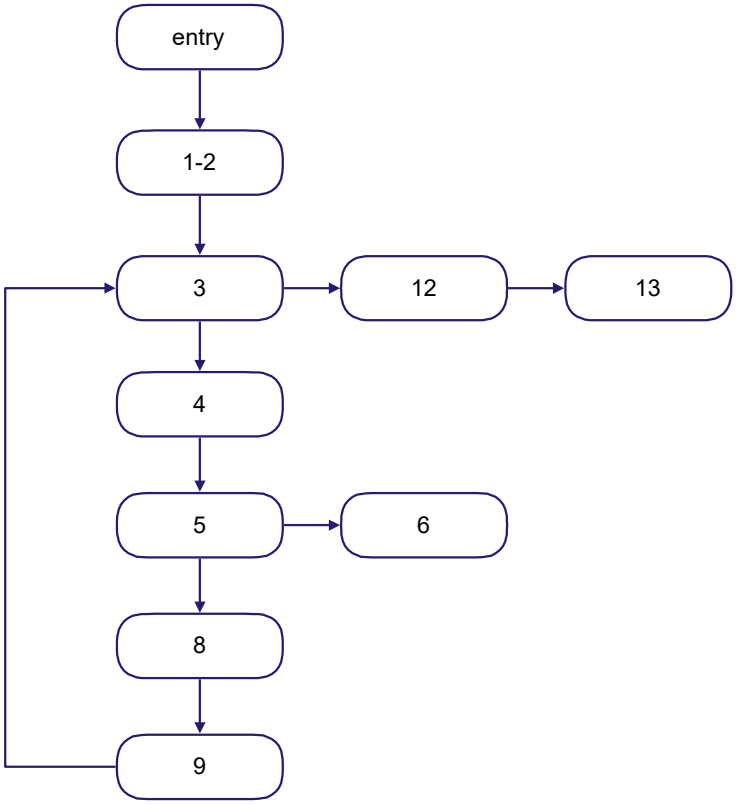


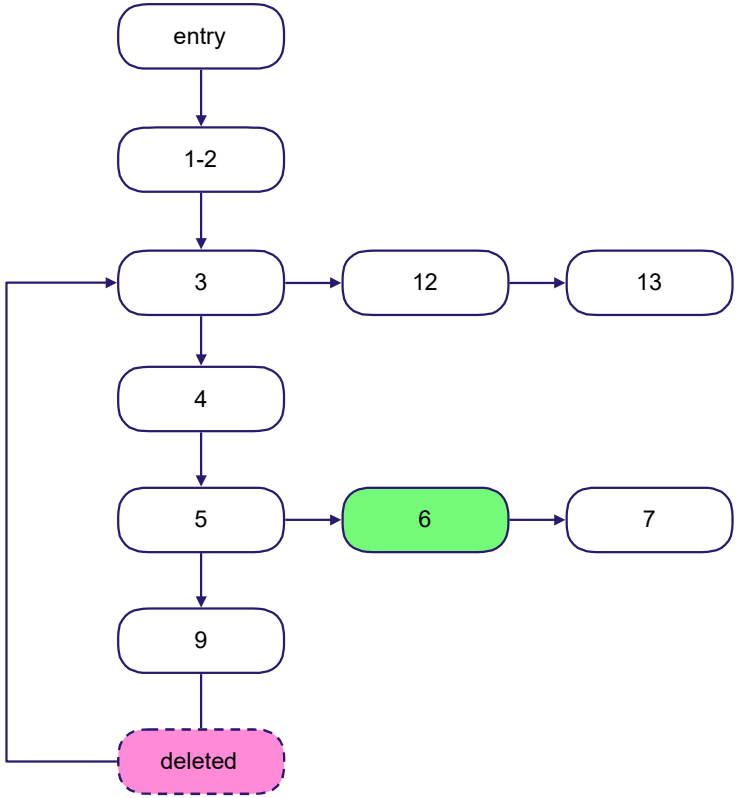


Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
t3	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 8), (8, 9), (9, 3), (3, 12), (12, 13)

An Average Example

Start with **selection = ∅**





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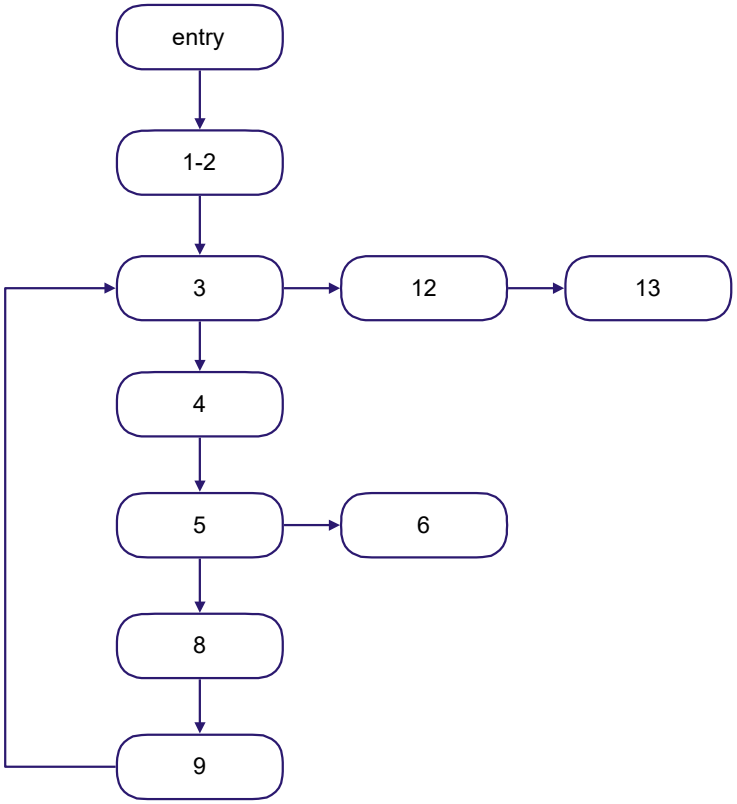
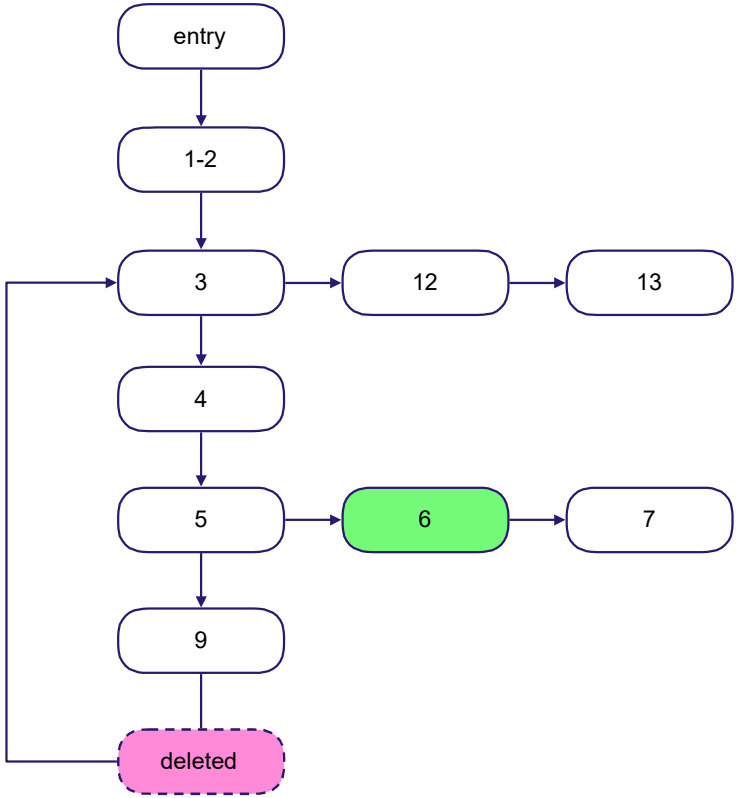
An Average Example





Start with **selection = ∅**

Traverse CFGs in parallel

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
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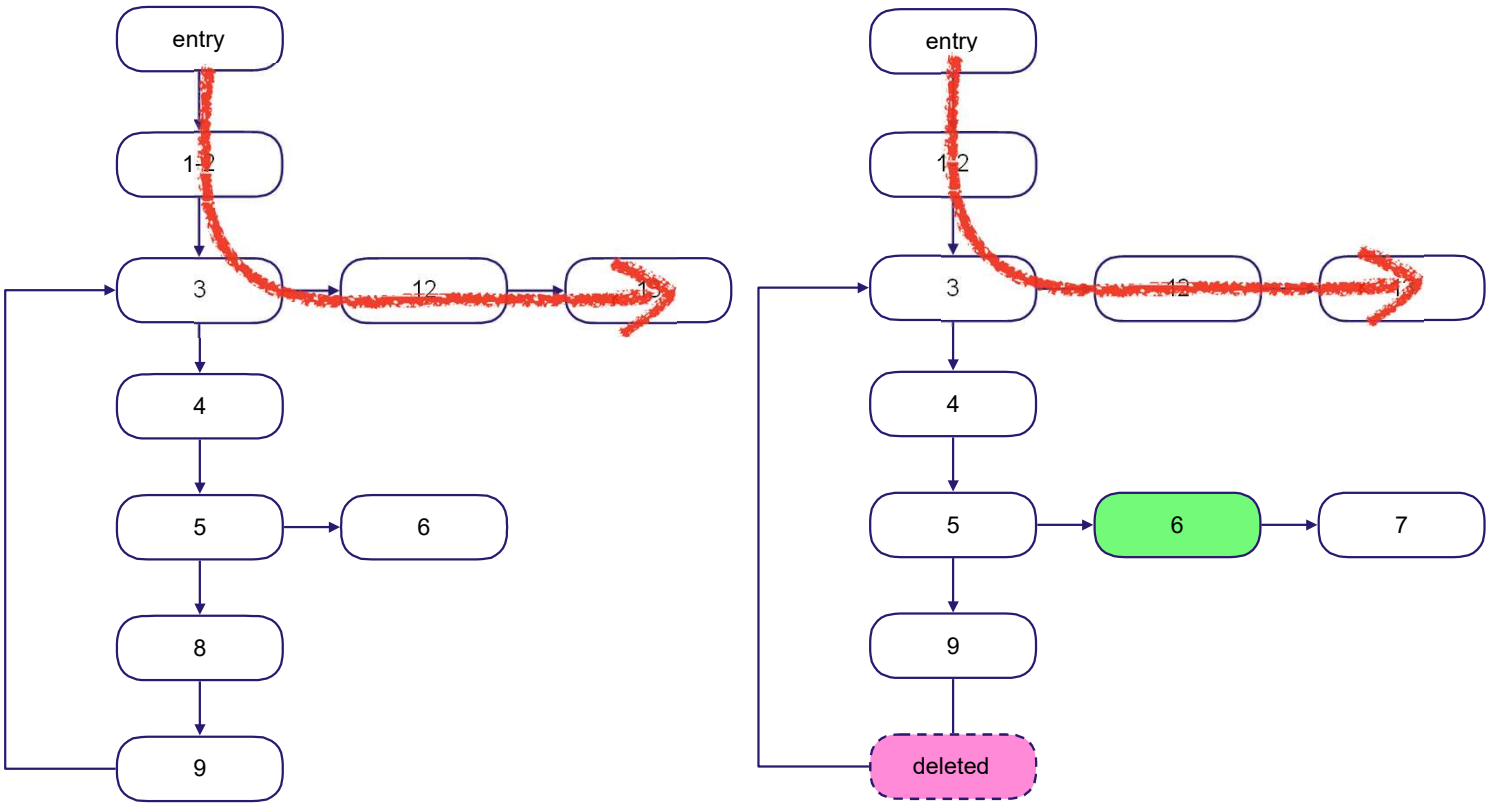
An Average Example



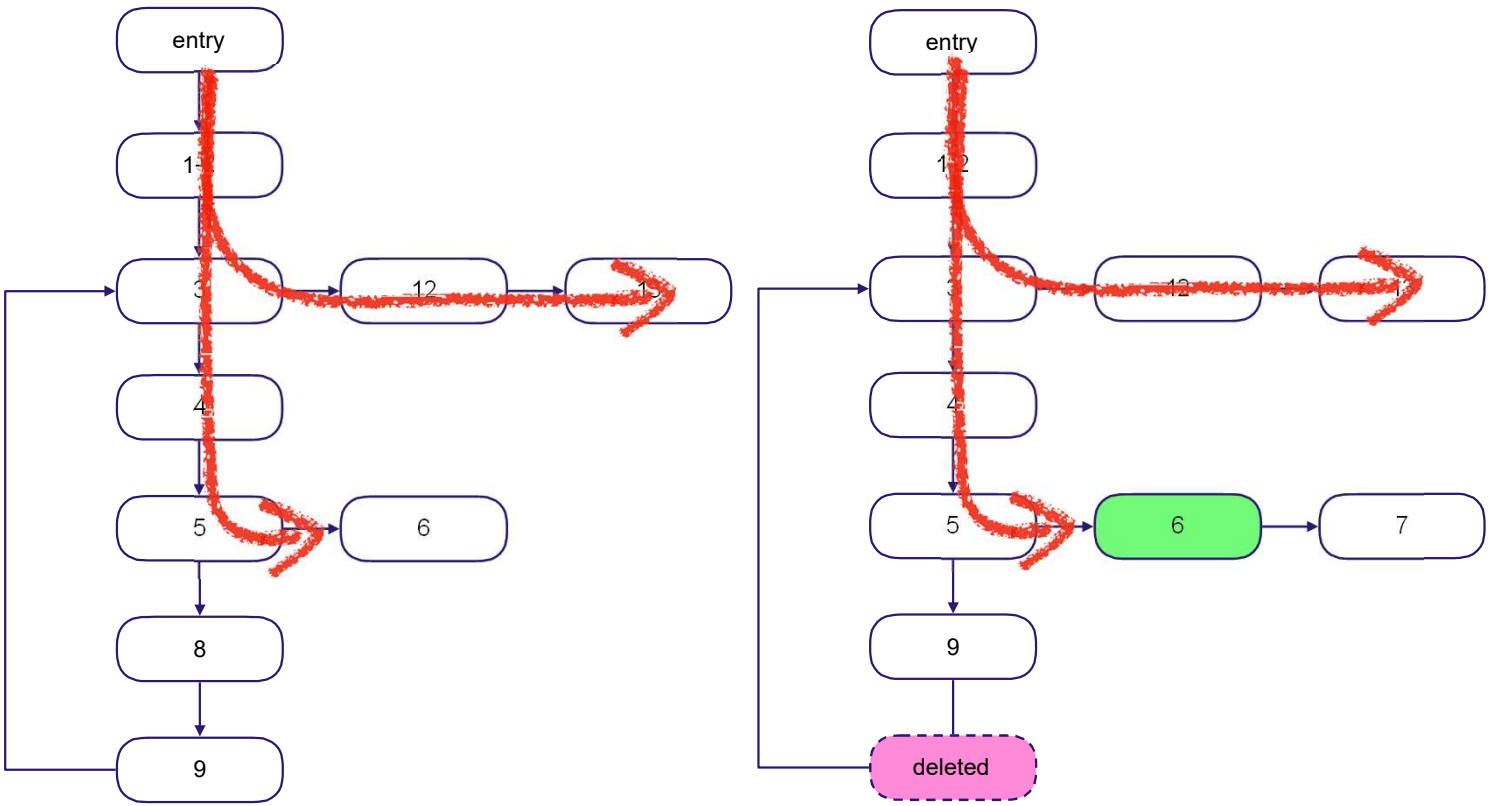
Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
t3	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 8), (8, 9), (9, 3), (3, 12), (12, 13)

An Average Example



Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

Once difference is found

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
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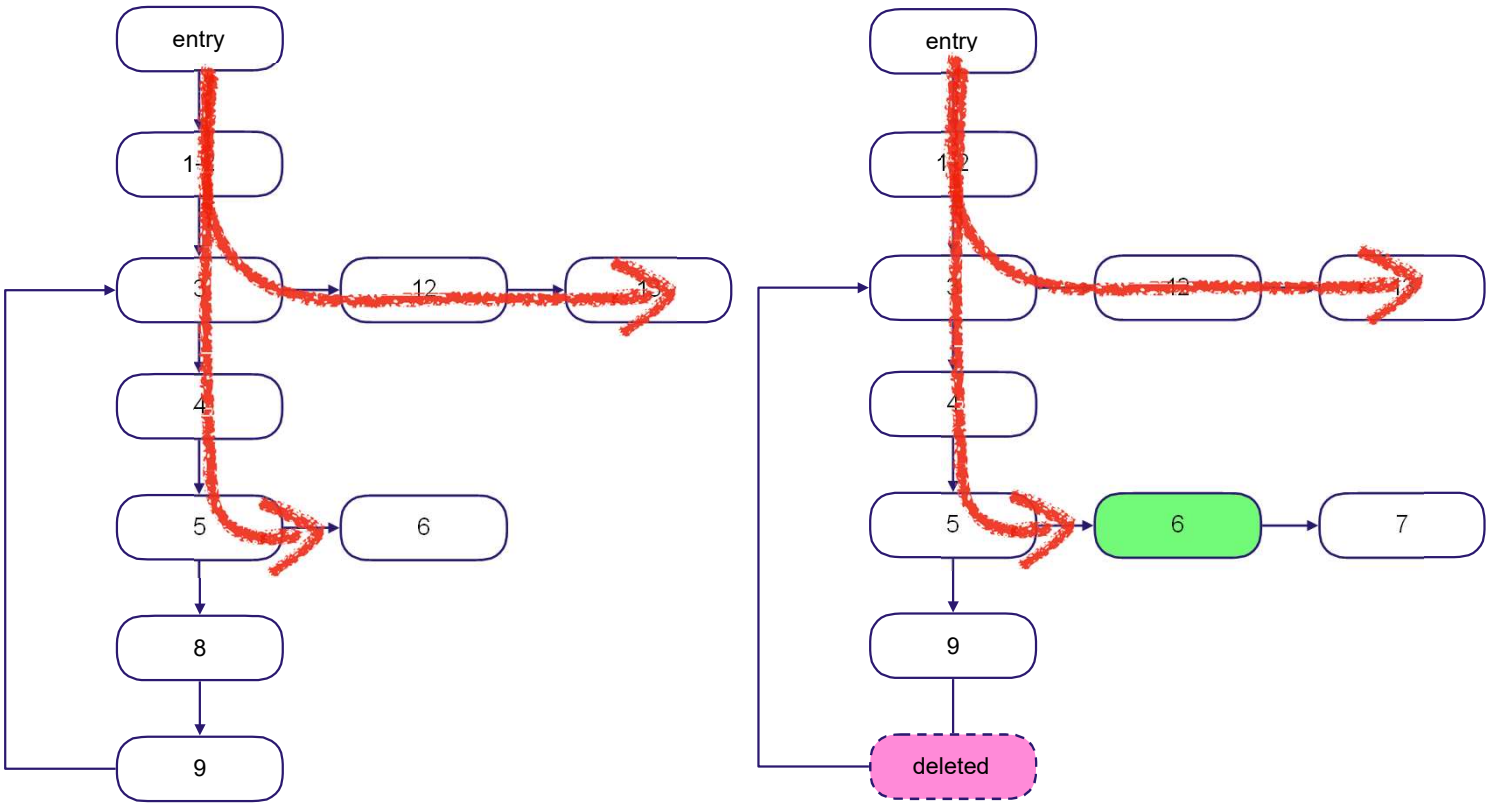
An Average Example



Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

Once difference is found

Add test to selection

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
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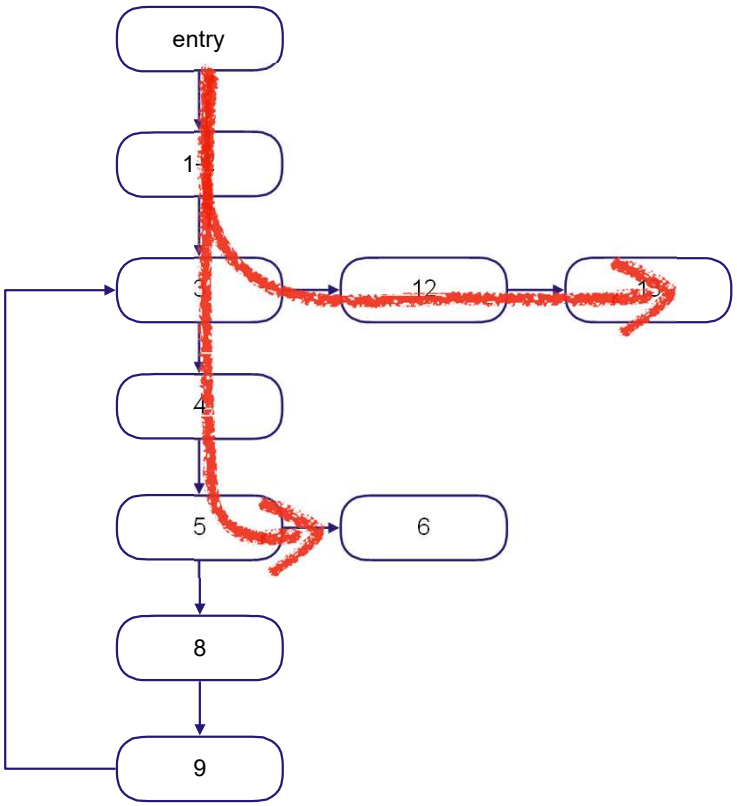
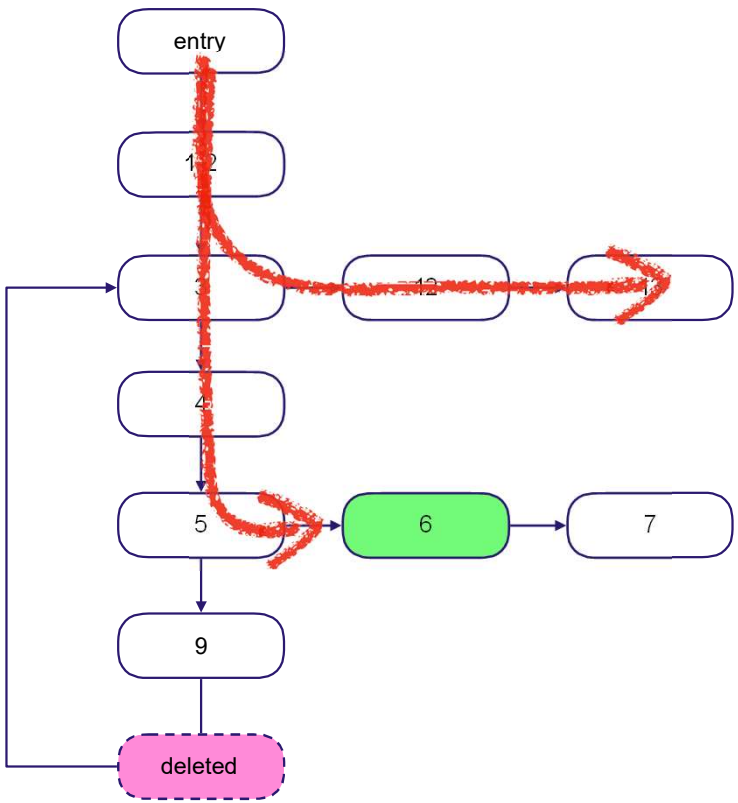




Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

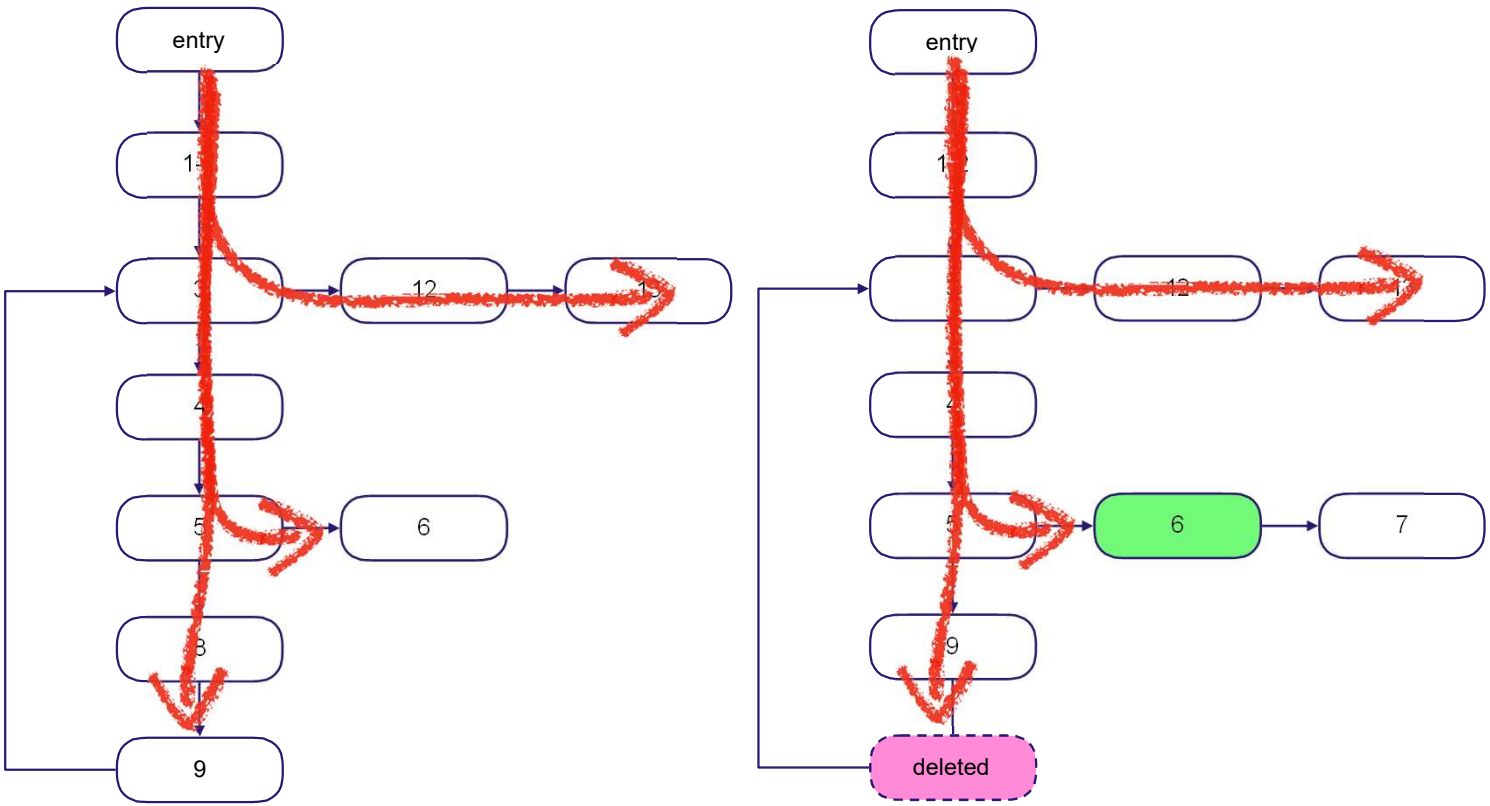
Once difference is found

Add test to selection

selection = { t2 }

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
t3	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 8), (8, 9), (9, 3), (3, 12), (12, 13)

An Average Example



Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

Once difference is found

Add test to selection

selection = { t2 }

Repeat recursively

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
t3	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 8), (8, 9), (9, 3), (3, 12), (12, 13)

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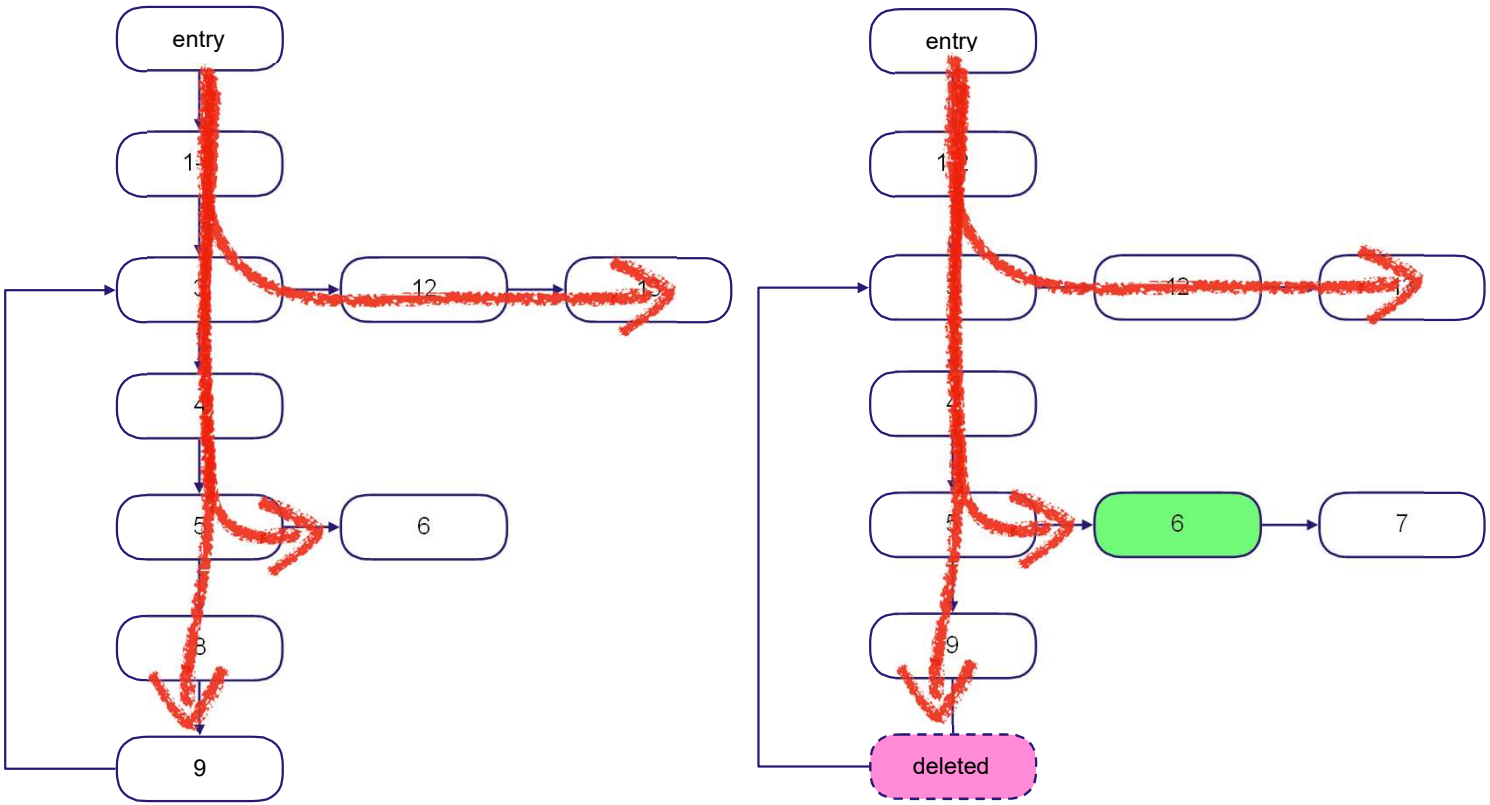

Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

Once difference is found

Add test to selection

selection = { t2 }

Repeat recursively

selection = { t2, t3 }

Test	Edges Traversed
t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
t2	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 6)
t3	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 8), (8, 9), (9, 3), (3, 12), (12, 13)

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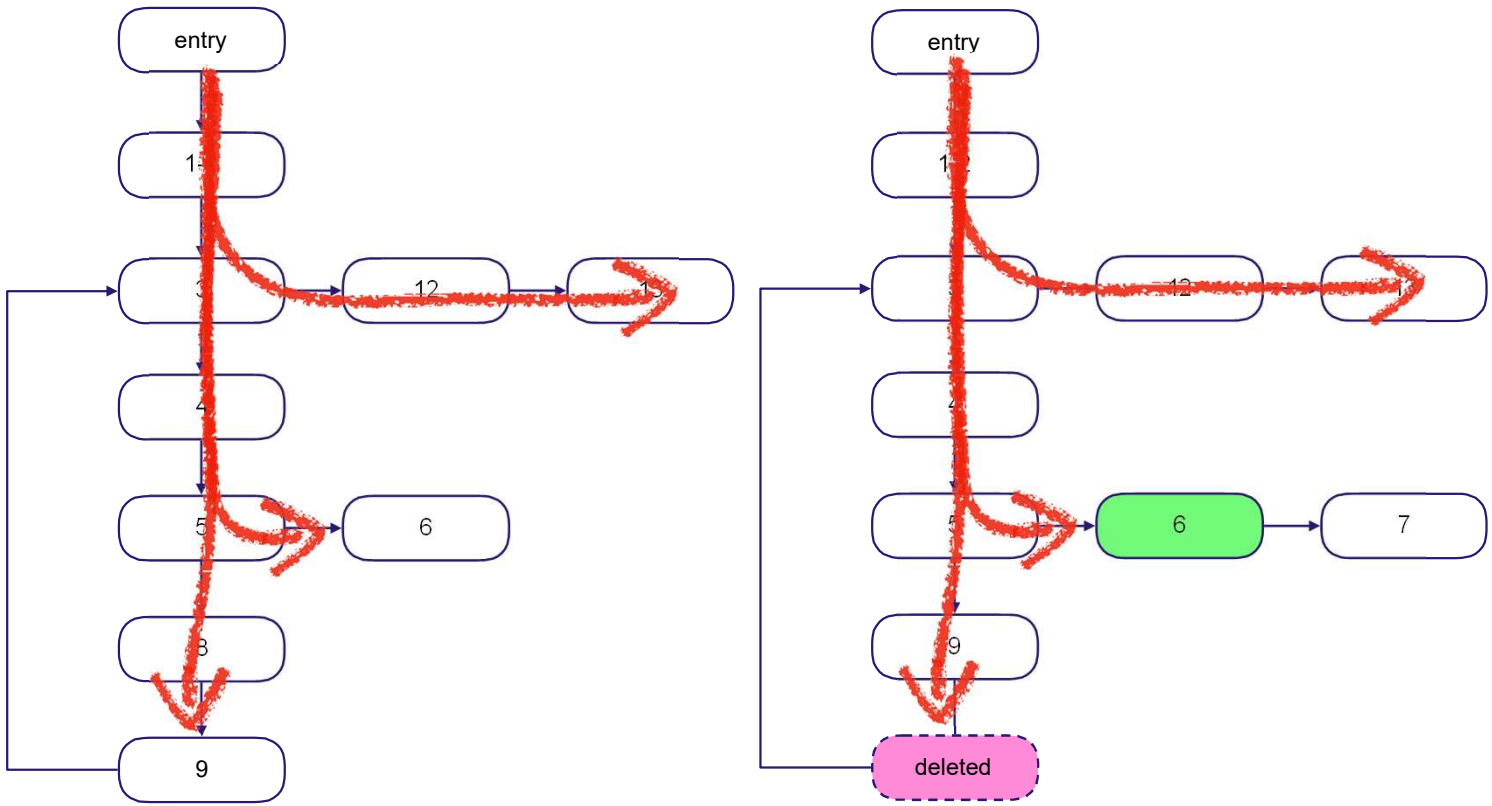

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t1	(entry, 1-2), (1-2, 3), (3, 12), (12, 13)
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t3	(entry, 1-2), (1-2, 3), (3, 4), (4, 5), (5, 8), (8, 9), (9, 3), (3, 12), (12, 13)

Start with **selection = ∅**

Traverse CFGs in parallel

To spot differences

Once difference is found

Add test to selection

selection = { t2 }

Repeat recursively

selection = { t2, t3 }

Complete traversal of CFGs

Research Highlights

Harrold MJ, Gupta R, Soffa ML. **A methodology for controlling the size of a test suite**. ACM Transactions on Software Engineering and Methodology (TOSEM) 2(3):270–285 (1993)

Rothermel G, Untch RH, Chu C, Harrold MJ. **Prioritizing test cases for regression testing**. IEEE Transactions on Software Engineering (TSE) 27(10):929-948 (2001)

Hadi Hemmati: **Chapter Four - Advances in Techniques for Test Prioritization**. Advances in Computing, vol 112: 185-221 (2019).

Rothermel G, Harrold MJ. **A safe, efficient regression test selection technique**. ACM Transactions on Software Engineering and Methodology (TOSEM) 6(2):173–210 (1997)

Summary

Multiple scenarios in which test suites will be used

- Continuous Integration helps with automation of test execution
- **Regression Testing** helps **find bugs** early

Benefits to minimisation, prioritisation and selection

- Simple heuristics – Can be expressed as **optimisation problems**
- Many different approaches, from simple greedy algorithms to multi-objective evolutionary algorithms